

Financial Frictions, Used Capital, and Misallocation

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Abstract. Standard macro-finance models treat capital as homogeneous. Financially constrained firms can therefore adjust only the quantity of capital they operate, not its vintage. This assumption is at odds with the data: young, small, and likely constrained firms use second-hand equipment disproportionately. I build a dynamic macro-finance model in which firms rent both new and used capital. Second-hand capital lets constrained firms operate closer to their unconstrained scale, even when they cannot meet the collateral requirements for new capital. It thereby reduces the dispersion of marginal revenue products of capital and weakens the dependence of entrepreneurship on initial wealth. I calibrate the model to Vietnamese firms and find that used capital mitigates the allocative and occupational distortions created by financial frictions.

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1. Introduction

Why is productivity so much lower in poor countries? Differences in physical and human capital explain only part of the large gap in output per worker across countries. The rest is often linked to total factor productivity. One reason is that resources do not always go to the firms that can use them best. Policies, weak institutions, and financial frictions can prevent productive firms from growing. This lowers aggregate productivity and helps explain lower productivity in developing countries (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009; Restuccia and Rogerson, 2017).

Capital misallocation is one source of these productivity losses (David and Venkateswaran, 2019; Bau and Matray, 2023). Financial frictions can prevent productive firms with little wealth from obtaining the capital they need. A large quantitative literature studies this mechanism in dynamic general equilibrium models with collateral constraints (Buera et al., 2011; Moll, 2014; Midrigan and Xu, 2014). A common feature of these models is that capital is homogeneous: a financially constrained firm can adjust the quantity of capital it operates, but not whether that capital is new or used. This omits an adjustment margin that matters in the data, since young and financially constrained firms rely disproportionately on used equipment (Eisfeldt and Rampini, 2007; Ma et al., 2022). How much does access to used capital change the aggregate productivity losses caused by financial frictions?

I address this question by introducing a choice between new and used capital into a dynamic general equilibrium model with collateral constraints and endogenous occupational choice. Firms rent both new and used capital from a financial intermediary, choosing the mix jointly with their operating scale. The model therefore links firms' use of used equipment to their productivity and financial position. It also allows second-hand capital to affect both the allocation of capital across operating firms and the decision to become an entrepreneur.

The central mechanism is a trade-off between the rental cost of used equipment and the collateral required to rent it. Used equipment may carry a higher rental cost per physical unit. Still, because its market value is lower, it absorbs less of a constrained entrepreneur's financing capacity per dollar of rental spending. A productive entrepreneur with little wealth may consequently rent used capital and operate at a scale that new capital alone would not finance. As wealth rises, the collateral constraint becomes less important and firms' choices converge toward the unconstrained mix. The same mechanism raises the value of entrepreneurship for productive but poor agents and thus weakens the dependence of occupational choice on initial wealth.

I quantify this mechanism using firm-level data on Vietnamese small and medium-sized enterprises. These data offer a rare opportunity to study this margin: the Vietnam SME Survey follows the same firms across six rounds from 2005 to 2015, and in every round it records the share of each firm's main equipment bought second-hand, as well as its assets, debt, and access to credit (CIEM et al., 2005–2015). The benchmark economy contains both new and used capital; the comparison economy removes second-hand capital. I calibrate the two separately, so that each reproduces the same five targets, including the debt-to-output ratio and the top-decile employment share; the benchmark additionally matches the decline in used-capital reliance with firm assets and the aggregate ratio of

new to used capital. I then remove the collateral constraint within each calibrated economy and recompute equilibrium prices, occupational choices, and stationary distributions. The comparison identifies both the contribution of second-hand capital and the cost of the collateral constraint.

Quantitatively, used capital has two effects. First, it improves how capital is allocated: relative to a single-vintage economy—one in which firms rent only new capital, calibrated on its own to the same five empirical targets—access to second-hand capital reduces the misallocation that collateral constraints generate by 39 percent. Second, it changes who runs a firm. Collateral constraints keep many talented but poor agents out of entrepreneurship, and access to second-hand capital repairs 23 percent of that lost entry, while entry by talented rich agents is nearly unchanged. Entrepreneurship therefore depends much less on initial wealth.

This paper contributes to the literature on factor misallocation and financial frictions, in which a constrained firm can adjust only the quantity of capital it operates.¹ This paper adds the choice of which kind of capital to operate, and that choice turns out to matter: access to second-hand capital removes a substantial part of the misallocation that collateral constraints generate in an economy without second-hand capital. The paper also relates to research on entrepreneurship and wealth.² I differ from that work in giving the constrained agent a second capital type rather than only a smaller scale, and I show that this changes who becomes an entrepreneur: because used equipment absorbs less financing capacity per dollar of rental spending, a talented agent can enter on less wealth than the collateral constraint would otherwise require.

The paper is closest to work on used capital and financing. [Eisfeldt and Rampini \(2007\)](#) show that the used share of capital expenditure falls with firm size and rises with measures of financial constraint, and [Ma et al. \(2022\)](#) document that young firms buy disproportionately old equipment. [Eisfeldt and Rampini \(2007\)](#) explain the pattern by the timing of payments, since used capital is cheap up front but costly to maintain later, while [Ma et al. \(2022\)](#) point to the smaller down payment that older equipment requires. [Rampini and Viswanathan \(2013\)](#) and [Rampini \(2019\)](#) establish the market-value collateral logic I use, in which a more durable asset requires a larger down payment; here durability varies within a single asset class that firms combine, so the response is a change in the capital mix rather than a move between classes. A parallel literature studies the market in which second-hand capital trades—irreversibility and capital specificity ([Lanteri, 2018](#)), constrained-efficient reallocation ([Lanteri and Rampini, 2023](#)), adverse selection over quality ([Bierdel et al., 2023](#)), and trade in whole firms ([Guntin and Kochen, 2026](#)). I take the supply of second-hand capital as given and ask instead what its availability does to misallocation and to entry in general equilibrium.

The rest of the paper is organized as follows. Section 2 reviews the related literature. Section 3

¹On the measurement of misallocation and its aggregate cost, see [Restuccia and Rogerson \(2008\)](#), [Hsieh and Klenow \(2009, 2014\)](#), [David and Venkateswaran \(2019\)](#), [Bau and Matray \(2023\)](#), and [Adamopoulos and Restuccia \(2014\)](#). On financial frictions in dynamic general equilibrium models with collateral constraints, see [Buera et al. \(2011\)](#), [Moll \(2014\)](#), [Midrigan and Xu \(2014\)](#), and [Chen et al. \(2023\)](#); for a survey, see [Buera et al. \(2015\)](#). On distortions and selection into production, see [Adamopoulos et al. \(2022\)](#).

²[Evans and Jovanovic \(1989\)](#), [Cagetti and De Nardi \(2006\)](#), [Fonseca et al. \(2007\)](#), [Buera and Shin \(2013\)](#), and [Tan \(2026\)](#).

describes the data. Section 4 presents the motivating evidence from the Vietnamese SME survey. Section 5 develops the model. Section 6 presents the calibration and quantitative results. Section 7 concludes.

2. Related Literature

This paper builds on four related literatures: misallocation, financial frictions, entrepreneurship and wealth, and used capital. It examines how firms' choice between new and used equipment changes the productivity losses and occupational decisions caused by collateral constraints.

Misallocation and its measurement. Restuccia and Rogerson (2008) show that distortions across heterogeneous producers can generate large aggregate productivity losses. Hsieh and Klenow (2009) quantify the potential gains from reallocating capital and labor across manufacturing plants in China and India. Restuccia and Rogerson (2017) survey this literature and emphasize the importance of identifying the specific frictions that cause misallocation.

David and Venkateswaran (2019) study why the return to capital differs across Chinese manufacturing firms. Adjustment costs and uncertainty explain only a small part of these differences. Most are linked to productivity or persistent differences across firms. Financial frictions and policies that affect firms differently by size may also matter, but the analysis cannot identify one single cause. Hsieh and Klenow (2014) show that plants in India and Mexico expand much less over their life cycle than US plants. Their model suggests that slower firm growth lowers manufacturing productivity. This motivates the study of barriers that prevent firms from expanding.

Adamopoulos et al. (2022) show that distortions in China affect both how resources are allocated across farmers and which workers remain in agriculture. In their model, this selection effect amplifies the productivity losses from misallocation.

My paper studies both effects in a model with collateral constraints. It considers how capital is allocated across operating firms and whether productive but poor people become entrepreneurs. I examine how access to used equipment changes these outcomes.

Financial frictions in quantitative general equilibrium. A large quantitative literature studies financial frictions in models with heterogeneous producers. Buera et al. (2011) show that financial frictions distort the allocation of capital and entrepreneurial talent, with larger effects in sectors that require a larger scale of operation. Saving can partly offset these constraints. Moll (2014) shows that the effectiveness of self-financing depends on the persistence of productivity: sufficiently persistent shocks can imply small steady-state losses but slow transitions. Buera and Shin (2013) show that financial frictions slow resource reallocation after reforms, generating gradual increases in investment and productivity. Midrigan and Xu (2014) find that capital misallocation among existing producers generates relatively small productivity losses, while distorted entry and technology adoption account for larger losses. Buera et al. (2015) provide a review of this literature.

Chen et al. (2023) show that managerial inputs can weaken the self-financing mechanism. As firms grow, their expenditure share on managers rises, reducing profit margins and slowing wealth

accumulation. Managerial inputs also increase capital demand. Together, these effects make financial constraints more persistent, including among large firms, and amplify productivity losses. My paper adds a choice between new and used equipment. Alongside saving and adjusting their scale, firms can change their capital mix in response to a collateral constraint. I quantify how this choice changes the constraint’s aggregate effects.

Entrepreneurship and wealth. The relationship between wealth and entrepreneurship is central to models of occupational choice with borrowing constraints. [Evans and Jovanovic \(1989\)](#) study how liquidity constraints affect entrepreneurial selection. [Cagetti and De Nardi \(2006\)](#) show how borrowing constraints and entrepreneurial saving jointly shape business ownership and the wealth distribution. [Fonseca et al. \(2007\)](#) show in a model that start-up costs can weaken the relationship between wealth and entrepreneurship and provide supporting cross-country evidence from older adults.

[Tan \(2026\)](#) studies how capital resale frictions affect entrepreneurial investment and the wealth distribution. When capital is illiquid, entrepreneurs face greater liquidity risk and lower collateral values, which discourages investment and worsens capital misallocation. His mechanism runs through the difficulty of reselling entrepreneurial capital: limited resale options raise the cost of adjusting investment and weaken the collateral backing of entrepreneurial assets, with consequences for both aggregate efficiency and the wealth distribution.

[Guntin and Kochen \(2026\)](#) study how trade in existing firms can alleviate financial constraints. They document that many US entrepreneurs purchased their business rather than founding it, and that buyers are wealthier than sellers on average. Following a sale, capital grows faster than output, reducing the firm’s average revenue product of capital. Their model shows how transferring ownership can improve capital allocation.

My paper connects occupational choice to firms’ choice between new and used equipment. In my model, used equipment absorbs less financing capacity per dollar of rental spending. Access to used capital can therefore make entrepreneurship more attractive to productive people with little wealth. I quantify how this choice changes the dependence of entrepreneurship on wealth.

Used capital and financing. [Eisfeldt and Rampini \(2007\)](#) study why financially constrained firms use second-hand capital. They show that the share of used capital in capital expenditures declines with firm size and is much higher among the smallest firms than among the largest. Used-capital investment is also associated with measures of financial constraints. Their mechanism works through the timing of payments: used equipment requires a smaller initial payment but higher subsequent maintenance costs, making it attractive to firms that are currently constrained. [Ma et al. \(2022\)](#) document a related pattern: young firms purchase disproportionately old equipment, with evidence consistent with financial constraints.

The financing advantages of different assets also depend on their durability and collateral value. [Rampini and Viswanathan \(2013\)](#) study how collateral constraints affect investment, financing, and leasing. [Rampini \(2019\)](#) shows that more durable assets require larger down payments because

durability raises their price by more than their collateral value. Financially constrained firms therefore prefer less durable and used assets. My model incorporates a choice between new and used capital into a general equilibrium setting where firms rent both types from a financial intermediary and agents choose between entrepreneurship and wage work.

Related work examines used-capital prices and resource allocation. [Lanteri \(2018\)](#) documents that capital reallocation and used-capital prices are procyclical. His model explains these patterns through firm-level capital specificity, which makes new and used investment goods imperfect substitutes. My model also treats new and used capital as imperfect substitutes. [Edgerton \(2011\)](#) estimates the elasticity of substitution between new and used equipment for farm and construction machinery and finds that the two are substitutes, but imperfect ones, providing an empirical benchmark for this parameter.

[Lanteri and Rampini \(2023\)](#) study the efficiency of capital reallocation when firms face collateral constraints. Used-capital prices affect both collateral values and the allocation of resources between buyers and sellers. [Bierdel et al. \(2023\)](#) study how asymmetric information about capital quality distorts trade in illiquid markets, with larger effects during crises. My paper examines how access to used capital changes the productivity losses and occupational choices caused by collateral constraints. I quantify these effects using Vietnamese firm-level panel data that combine information on second-hand equipment use with firms' financial positions.

3. Data

3.1. The Vietnam SME Survey

I use the Vietnamese Small and Medium Enterprise Survey ([CIEM et al., 2005–2015](#)), a biennial firm-level panel of small and medium-sized manufacturing enterprises run by CIEM, ILSSA, the University of Copenhagen, and UNU-WIDER. Owners or managers are interviewed in person in ten provinces, and the same firms are revisited every two years. The six rounds from 2005 to 2015 cover 5,131 firms and 15,811 firm-wave observations.

For each firm the questionnaire records the age of its main equipment, the share acquired new, second-hand, or self-constructed, whether it is owned or rented, and how it was financed, alongside total assets, formal and informal debt, loan applications, whether the firm had problems obtaining a loan, and, for non-applicants, why not.

Three features make the survey suited to the question. The firms are small, mostly household-owned, and largely outside formal credit, which is the population the model describes. Used equipment is common rather than marginal. I can observe a firm's reliance on used equipment change with its own financial position.

The survey is designed to represent non-state manufacturing firms in the ten provinces: Ha Noi, Phu Tho, Ha Tay, Hai Phong, Nghe An, Quang Nam, Khanh Hoa, Lam Dong, Ho Chi Minh City and Long An. Together they held about 30 percent of Vietnam's non-state manufacturing firms in 2005. The sampling frame combines the 2002 Establishment Census, which lists household businesses, with the Industrial Survey, which lists firms registered under the Enterprise Law. Districts are

Table 1. The Vietnam SME Survey, round by round

	2005	2007	2009	2011	2013	2015	Pooled
<i>Panel A. The sample</i>							
Firms interviewed	2,819	2,631	2,659	2,512	2,542	2,648	15,811
of which not in the previous round	—	339	549	459	484	530	—
Re-interviewed two years later (%)	81.3	80.2	77.2	81.9	83.3	—	80.8
Household businesses (%)	68.3	68.1	65.2	64.4	62.7	62.9	65.3
Full-time workers, median	6	6	6	5	5	4	5
Firm age in years, median	10	12	11	12	13	15	12
No outstanding debt (%)	40.5	45.0	40.7	49.6	49.0	62.0	47.7
Applied for a formal loan (%)	39.3	36.8	37.5	29.5	26.0	24.7	32.4
<i>Panel B. The used-capital item</i>							
Firms answering the item	2,524	2,420	2,467	2,364	2,401	2,515	14,691
Any second-hand equipment (%)	46.1	35.5	37.2	33.1	36.7	26.6	35.9
All equipment second-hand (%)	24.2	15.5	13.9	11.6	11.4	7.1	14.0
Used share, mean over firms (%)	35.6	24.6	25.1	22.3	23.8	15.7	24.5

Notes: Author’s calculations from the six enterprise rounds of the Vietnam SME Survey ([CIEM et al., 2005–2015](#)). The pooled column counts firm rounds; the panel follows 5,131 firms. Firms not in the previous round are mostly new to the survey; a few return after missing a round. Firms, workers and debt refer to the end of the year before the interview. Formal loan applications cover the period since the previous round (since 2002 in the 2005 round). Panel B covers firms that use machinery and whose new, second-hand and self-constructed percentages add up to 100.

drawn with probability proportional to size, and firms are drawn at random within each district. Because informal firms appear in neither list, enumerators also interview informal firms that they find operating in the selected districts. State-owned and foreign-invested firms are not in the frame ([Berkel et al., 2020](#)).

The survey follows the same firms over time. Every two years the team returns to the firms of the previous round, and firms that have closed are replaced at random from the updated frame. Of the 5,131 firms, 1,402 are interviewed once, 1,031 twice, 807 three times, 536 four times, 348 five times and 1,007 in all six rounds. Of the 2,819 firms interviewed in 2005, 1,035 (36.7 percent) are interviewed again in 2015. The team also follows up on firms it cannot find, and [Berkel et al. \(2020\)](#) conclude that most of them had closed rather than refused or moved. They put the annual exit rate at 9 to 10 percent, in line with the 9.8 percent that I compute from the survey and use as a calibration target in Section 6.

Table 1 describes each round. Between 2,512 and 2,819 firms are interviewed in each round, and four in five of them are interviewed again two years later. The firms are small: the median firm has four to six full-time workers, and about two thirds are household businesses. Panel B shows that second-hand equipment is common in every round. Between a quarter and a half of the firms operate some second-hand equipment, and those that do obtained, on average, about two thirds of their main equipment second-hand. The share of firms with any second-hand equipment falls from 46 percent in 2005 to 27 percent in 2015, while the panel ages: the median firm is 10 years old in 2005 and 15 in 2015.

Table 2. Variables used from the survey

Variable	Definition	Rounds
Used share	Percentage of the firm’s main equipment and machinery obtained second-hand. Firms that use machinery and whose new, second-hand and self-constructed percentages add up to 100.	All
Any / all second-hand	Used share above 0 / equal to 100.	All
Equipment age	Percentage of main equipment in five age bands. The mean uses the band midpoints 1.5, 4, 8, 15.5 and 25 years.	All
Firm age	Survey year minus the year the firm was established. For firms observed in multiple rounds, the establishment year is fixed at the firm’s most frequent reported answer.	All
Full-time workers	Full-time workers at the end of the previous year; classes 1–9, 10–49 and 50 or more.	All
Total assets	Physical and financial assets at the end of the previous year; book value in 2005 and 2007, market value from 2009.	All
Debt	Total outstanding debt at the end of the previous year.	All
Household business	Legal status is household business.	All
Formal loan application	Applied for a bank loan or other formal credit since the previous round.	All
Loan problems	Applicant had problems obtaining the loan, and the main cause.	2011–15
Reason for not applying	Main reason given by firms that did not apply for a formal loan.	2011–15

Notes: Workers, assets and debt come from the economic-accounts module, which measures outcomes at the end of the year before the interview. “All” denotes the six rounds: 2005, 2007, 2009, 2011, 2013, and 2015.

3.2. The Used-Capital Item

The 2013 questionnaire asks: “What percentage of your main equipment/machinery did you get: new; used/second hand; self-constructed?” Every round asks the same question; the 2007 and 2009 files record the answers as fractions, which I rescale. The question is meant for firms that use machinery, which are 89.5 percent of firms in 2005 and 95.0 percent in 2015, and it is answered in 95 percent of firm-rounds. In this section I use the 14,691 firm-rounds from firms that use machinery and whose three percentages add up to 100. Section 4 uses every firm that answers the item. Pooled over rounds, 60.5 percent of these firms obtained all their main equipment new, 14.0 percent obtained all of it second-hand, and 20.3 percent report both new and second-hand equipment. The remaining 5.2 percent hold some self-constructed equipment and do not mix new with second-hand equipment. The question does not say whether the percentages refer to the number or the value of machines; Section 6 explains how the calibration reads them.

A separate question asks for the age of the same equipment, in five bands. The mean age of the main equipment is 6.2 years in firms with no second-hand equipment and 9.0 years in firms whose equipment is all second-hand. The share older than ten years rises from 12.5 to 26.3 percent between these two groups, and the share younger than three years falls from 22.1 to 7.5 percent.

3.3. Changes within Firms

The panel follows the same firms from one survey round to the next, two years later. Table 3 pools the five pairs of consecutive rounds, from 2005–2007 to 2013–2015, which gives 8,713 observations from 3,333 firms. Each row starts from the firms’ equipment in one round and shows their equipment in the next round. One firm in three (33.2 percent) is in a different group two years later. Firms with only new equipment usually keep it: 81.8 percent still have only new equipment two years later. Firms with only second-hand equipment often move away from it: two years later, 40.5 percent have only new equipment and 37.6 percent still have only second-hand equipment. This move to only new equipment happens in every pair of rounds, for between 31 and 47 percent of these firms, and 510 firms make it at least once. This within-firm variation is what identifies the fixed-effects estimates in Section 4.

Table 3. How firms’ equipment changes over two years

Equipment in one round	Equipment in the next round (% of firms)			Observations
	Only new	New and used	Only second-hand	
Only new	81.8	12.3	5.9	5,452
New and used	44.8	44.6	10.6	1,919
Only second-hand	40.5	21.9	37.6	1,342

Notes: Each row follows the same firms to the next survey round, two years later, and adds up to 100. For example, of the firms with only second-hand equipment in one round, 40.5 percent have only new equipment in the next. The table pools the pairs of rounds 2005–2007, 2007–2009, 2009–2011, 2011–2013 and 2013–2015. An observation is a firm that answers the equipment question in both rounds of a pair. The 925 pairs in which the firm reports self-built equipment without both new and second-hand equipment are left out. Author’s calculations from the Vietnam SME Survey (CIEM et al., 2005–2015).

4. Evidence on Used Capital and Financing Frictions

This section uses the panel described in Section 3 to ask whether a firm’s reliance on second-hand equipment falls as its own financial position improves.

Table 4 reports the used-capital share by firm age. Second-hand equipment accounts for 31.5 percent of a typical young firm’s main equipment, against 21.6 percent for firms aged 11 years or more. Similar gradients appear in richer economies (Eisfeldt and Rampini, 2007; Ma et al., 2022). A cross-section cannot tell whether the gradient reflects financing constraints or permanent firm differences. Proposition 2 predicts that within a firm, the used share should fall as collateral accumulates; Section 6.1 shows μ falling with wealth in the model.

I estimate

$$\text{UsedShare}_{it} = \chi_1 \text{ZeroDebt}_{it} + \chi_2 (\text{ZeroDebt}_{it} \times X_{it}) + \chi_3 X_{it} + \mathbf{\Gamma}' \mathbf{Z}_{it} + \lambda_t + \nu_i + \varepsilon_{it}, \quad (1)$$

where UsedShare_{it} is used capital as a percentage of firm i ’s main equipment and $\mathbf{Z}_{it}$ holds two employment-size indicators. Table 5 reports the estimates. Column (1) sets X_{it} to centered log

assets; column (2) sets X_{it} to a young-firm indicator and additionally controls for log assets. Firm and year fixed effects absorb permanent firm characteristics and common round shocks; identification comes from within-firm changes in debt status, assets, and age. Zero debt marks firms that finance themselves without borrowing, not the levered constrained firms in the model. Because some zero-debt firms simply choose not to borrow, χ_2 is likely too small rather than too large. Firms' direct reports of credit problems, discussed below, confirm the pattern.

Table 4. Used-capital share by firm age

Firm age	Share (%)	Observations
Young (0–5 years)	31.5	2,447
6–10 years	26.0	3,854
11 years or more	21.6	8,610

Notes: Firm-weighted averages over 14,911 firm-wave observations with a usable start year. Author's calculations from the pooled 2005–2015 waves of the Vietnam SME Survey (CIEM et al., 2005–2015).

The estimates match the collateral prediction. For zero-debt firms, the used share falls 1.07 percentage points more per log point of assets than for firms with debt (column 1), and the young-firm premium is 3.9 points larger, a total of 4.6 points ($0.755 + 3.885$ in column 2). The effects are modest relative to a mean used share of 24.5 percent, as fixed effects remove level differences between firms; the point is the sign and the within-firm pattern.³ The evidence is descriptive rather than causal.

The survey also records whether a firm that applied for a formal loan had problems obtaining it, which is the closest the data come to a direct report of a binding constraint. Among applicants in 2013, those reporting problems hold 11.7 percentage points more used capital (s.e. 3.10, $N = 644$); pooling the 2011–2015 rounds gives 6.8 percentage points (s.e. 1.82, $N = 2,010$). Results are similar with controls. The sample is restricted to applicants and is therefore selected, and the reports are the firm's own, so these estimates are not a substitute for the panel specification in (1). They do, however, point the same way as the zero-debt interactions and do so using a measure of credit access rather than a proxy for it.

Two points to note. First, there is no unconditional within-firm age gradient. In column (2), *Young* (0.755, s.e. 1.658) is the effect of being young for firms *with debt*; for zero-debt firms the within-firm premium is 4.6 percentage points ($p = 0.027$). The result is therefore the interaction with financing, not age alone. Second, zero debt marks self-financing rather than collateral rationing. Among firms that did not apply for a formal loan, 54.1 percent say they did not need one and a further 18.5 percent do not wish to incur debt; 11.8 percent report an interest rate they consider too high, 8.4 percent the difficulty of the process, 2.6 percent inadequate collateral, and 1.0 percent existing debt (pooled 2011–2015, $N = 5,618$). Stated reasons understate rationing, since a firm that

³Robustness checks on column (2): raw reported age gives 5.345 (2.378); firms in ≥ 3 rounds give 4.801 (2.439).

Table 5. Financing frictions and used capital

<i>Dependent variable:</i> used-capital share (percentage points)		
	(1) Asset interaction	(2) Young-firm interaction
Zero debt $\times$ log assets	−1.067** (0.462)	—
Zero debt $\times$ young	—	3.885* (2.340)
Zero debt	−0.482 (0.836)	−1.046 (0.872)
Log assets	−0.260 (0.493)	−0.679 (0.453)
Young (age ≤ 5)	—	0.755 (1.658)
Observations	13,492	13,403
Firms	3,551	3,549
Within R^2	0.036	0.036
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Size controls	Yes	Yes

Notes: Standard errors clustered by firm in parentheses. Log assets is centered at the sample mean. Size controls are indicators for 10–49 and 50 or more workers. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

expects to be refused does not apply and records that it did not need credit; among firms that did apply, 24.1 percent report problems obtaining the loan in 2013. The applicant evidence above is therefore the sharper test. Taken together, reliance on second-hand equipment falls with a firm’s own assets among plausibly constrained firms. Appendix A uses the same credit items in all six rounds, adds non-applicants who were put off by collateral or procedures, and compares each firm with itself. In the rounds when a firm is constrained, it holds 1.7 percentage points more of its main equipment second-hand, and when the problem was a lack of collateral, the used share rises by 6.0 points and the new share falls by 6.6.

5. Model

I develop a stationary general equilibrium model with endogenous occupational choice and a competitive rental market for new and used capital. Agents save through deposits with a competitive financial intermediary that owns the physical capital stock and rents capital services to entrepreneurs. All agents, including the intermediary, are price takers, and all rental rates and asset prices are equilibrium objects. The model departs from a standard collateral-constraint economy in one respect: firms rent both new and used capital, and the collateral constraint is written on the combined market value of the two. Throughout, capital of type $j \in \{n, u\}$ has price p_j and user cost R_j per dollar of its market value, so the rental payment per physical unit is $p_j R_j$.

Notation. Each symbol below has a single meaning throughout the paper. $G(a, z, y)$ is the stationary distribution of agents; the entry shortfall of poor talented agents is written $\mathcal{S}$ (Section 6.3). L is labor in efficiency units; the TFP loss from the collateral constraint is written $\mathcal{M}$ (Section 6.2). D is aggregate deposits held with the intermediary; a worker's labor income is written $I_w(y)$, and aggregate external finance is written B^{ext} . Roman letters (G, L, D) denote model objects; calligraphic letters ($\mathcal{M}, \mathcal{S}$) denote the two loss measures used to summarize the quantitative results, and $\mathcal{E}, \mathcal{W}$ denote the sets of entrepreneurs and workers.

Scope. The supply of used capital here is deliberately mechanical. New capital becomes used at the constant rate ζ , used capital breaks at the rate κ , and the intermediary prices both at zero profit, so no firm chooses when to sell a machine and no buyer worries about what a machine on offer is worth. Three frictions that a richer market for second-hand capital would carry are therefore absent: the irreversibility and capital specificity that make used-capital prices procyclical (Lanteri, 2018), the pecuniary externality that operates through those prices when buyers are collateral-constrained (Lanteri and Rampini, 2023), and the adverse selection that thins trade when quality is private information (Bierdel et al., 2023). The omission is a deliberate one. Each of those frictions raises the effective cost of obtaining used equipment, so a model that included them would deliver a weaker version of the relief quantified below, not a different sign: the mechanism requires only that a rental dollar spent on used capital absorb less financing capacity than a rental dollar spent on new, which Proposition 2 shows is a statement about user costs alone. The stationary environment is also silent about cyclical reallocation, which is where those frictions do most of their work.

5.1. Environment and Timing

Time is discrete and there is no aggregate uncertainty. A unit measure of infinitely lived agents is indexed by the individual state (a, z, y) , where a is financial wealth, z is log entrepreneurial productivity, and y is worker ability. Preferences are

$$U_0 = E_0 \sum_{t=0}^{\infty} \beta^t u(c_t), \quad u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}, \quad (2)$$

with discount factor $\beta \in (0, 1)$ and risk aversion σ .

5.2. Technology and Idiosyncratic States

Entrepreneurial technology. An entrepreneur with productivity z hires L efficiency units of labor and rents new and used capital, K_n and K_u , to produce

$$Y = \exp(z) \left(L^\alpha K^{1-\alpha} \right)^\eta, \quad \eta < 1, \quad (3)$$

where α is the labor share within the variable-input bundle and $\eta < 1$ governs the span of control, as in Lucas (1978), so that optimal firm scale is finite and increasing in productivity. New and used

capital are imperfect substitutes, with effective capital given by the constant-elasticity aggregate

$$K = \left[\phi^{1/\gamma} K_n^{(\gamma-1)/\gamma} + (1-\phi)^{1/\gamma} K_u^{(\gamma-1)/\gamma} \right]^{\gamma/(\gamma-1)}, \quad (4)$$

where $\gamma > 1$ is the elasticity of substitution between new and used capital and ϕ their relative weight. Entrepreneurial productivity is persistent:

$$z' = \begin{cases} z, & \text{with probability } \omega, \\ \tilde{z}', \quad \tilde{z}' \sim F_z, & \text{with probability } 1 - \omega, \end{cases} \quad (5)$$

so ω governs the persistence of productivity. The induced distribution of productivity *levels*, $\exp(z)$, has a Pareto right tail with parameter η_p .

Workers. A worker with ability y supplies y efficiency units inelastically and earns labor income

$$I_w(y) = Wy, \quad (6)$$

where W is the wage per efficiency unit. Ability follows

$$\log y' = \rho_y \log y + \sigma_y \varepsilon'_y, \quad \varepsilon'_y \sim N(0, 1), \quad (7)$$

with stationary standard deviation $\sigma_y / \sqrt{1 - \rho_y^2}$. Productivity and ability evolve independently.

5.3. Individual Decisions and Occupational Choice

Entrepreneurial profits. Conditional on operating a firm, an agent with wealth a and productivity z chooses labor and both new and used capital to maximize operating profits,

$$\pi(a, z) = \max_{K_n, K_u, L \geq 0} \left\{ \exp(z) \left(L^\alpha K^{1-\alpha} \right)^\eta - WL - p_n R_n K_n - p_u R_u K_u \right\}, \quad (8)$$

subject to K in (4) and the financing constraint

$$p_n K_n + p_u K_u \leq \theta a, \quad (9)$$

where a is the entrepreneur's asset holdings used as collateral and $\theta \geq 1$ governs the stringency of the collateral constraint, with a smaller θ indicating a tighter constraint. New and used capital each absorb financing capacity equal to their own market value.

Consumption and saving. The value conditional on entrepreneurship is

$$V_e(a, z, y) = \max_{\substack{c \geq 0 \\ a' \geq 0}} \{ u(c) + \beta E[\Omega(a', z', y') \mid z, y] \}, \quad (10)$$

subject to

$$c + a' = \pi(a, z) + (1 + r)a, \quad (11)$$

and the value conditional on wage work is

$$V_w(a, z, y) = \max_{\substack{c \geq 0 \\ a' \geq 0}} \{u(c) + \beta E[\Omega(a', z', y') \mid z, y]\}, \quad (12)$$

subject to

$$c + a' = I_w(y) + (1 + r)a. \quad (13)$$

Deposits are non-negative: agents cannot borrow against future labor income, and the only credit in the economy is the intermediated capital limited by (9).

Occupational choice. The value available at (a, z, y) is

$$\Omega(a, z, y) = \max \{V_e(a, z, y), V_w(a, z, y)\}, \quad (14)$$

and the agent operates a firm if

$$V_e(a, z, y) \geq V_w(a, z, y) \quad (15)$$

and supplies labor otherwise. These occupational flows, together with the resulting composition of the entrepreneurial sector, are equilibrium outcomes of (15).

5.4. The Financial Intermediary and Capital Prices

A competitive financial intermediary collects deposits at the risk-free rate r , owns the physical capital stock, and rents capital services to entrepreneurs. It earns zero profit. The final good is the numeraire and one unit of it becomes one unit of new capital, so $p_n = 1$ throughout. The stocks evolve as

$$K_{n,t+1}^s = (1 - \delta - \zeta)K_{n,t}^s + I_t, \quad (16)$$

$$K_{u,t+1}^s = (1 - \delta - \kappa)K_{u,t}^s + \zeta K_{n,t}^s, \quad (17)$$

where I_t is investment, δ is baseline depreciation, $\zeta \in (0, 1)$ is the fraction of new capital that becomes used each period, and κ is the fraction of used capital that breaks and is hence valueless, with $\delta + \zeta < 1$ and $\delta + \kappa < 1$. Used capital is formed from new capital that becomes used, so it is in positive supply without absorbing investment of its own. Write p_u for the price of used capital relative to new, so that $p_u R_u$ is the rental payment per physical unit.

Proposition 1 (Capital prices). *Let the intermediary discount at r , take prices as given, and choose investment to maximize the present value of rental income net of investment subject to (16)–(17), where new capital rents at $p_n R_n$ and used capital at $p_u R_u$ per physical unit. Then a stationary*

solution with $K_n^s, K_u^s \in (0, \infty)$ requires

$$R_n = r + \delta + \zeta \left(1 - \frac{p_u R_u}{r + \delta + \kappa} \right). \quad (18)$$

The proof is in Appendix B.

Only the product $p_u R_u$ appears in (18), and only that product enters a firm's rental bill, so the model determines the rental payment on used capital rather than its two factors. Define

$$R_u = r + \delta + \kappa, \quad (19)$$

the user cost of used capital: one dollar held in used capital must earn the deposit rate, cover depreciation, and cover the rate at which the machine breaks. With R_u so defined, (18) determines the price of used capital relative to new,

$$p_u = p_n - \frac{R_n - r - \delta}{\zeta}, \quad (20)$$

and (18) can be written $R_n = r + \delta + \zeta(p_n - p_u)$. The last term is the net cost of aging: with probability ζ a unit priced at p_n is repriced at p_u . At $\zeta = 0$ it collapses to $R_n = r + \delta$, which is the case in the economy without used capital.

This split would be immaterial if used capital merely produced. It matters here because the financing constraint (9) is written on market value, which makes p_u enter the entrepreneur's problem on its own. The split is not identified by rental payments, which involve only the product $p_u R_u$; it is pinned by requiring p_u to be the price at which a used unit trades and is carried on the intermediary's balance sheet (21), which is the price the collateral constraint counts. The mechanism of the paper originates in it: breakage raises R_u through (19), making used capital expensive to rent, while the price at which its rental stream capitalizes is determined separately by (20), making it cheap in collateral capacity.

Deposits are the intermediary's only liability and capital its only asset, so

$$D = p_n K_n^s + p_u K_u^s. \quad (21)$$

Given (21), the pricing condition implies zero profit, as Appendix B shows. The intermediary is a veil: the economy behaves as if households held the capital stock directly and priced it at (p_n, p_u) .

In a stationary equilibrium the used stock is constant, so

$$\zeta K_n^s = (\delta + \kappa) K_u^s, \quad \text{hence} \quad \frac{K_u^s}{K_n^s} = \frac{\zeta}{\delta + \kappa}. \quad (22)$$

This ratio is fixed by (ζ, δ, κ) and does not respond to prices; the levels of both stocks scale with aggregate wealth, as (27) makes explicit. Three prices clear three markets: the wage clears the labor market, the deposit rate clears the new-capital market through (21), and R_n adjusts until

the demand ratio matches (22). By (20) a higher R_n lowers p_u , which tilts demand toward used capital through both the rental payment and the collateral a unit absorbs, so used-capital clearing is automatic.

5.5. Aggregation and Stationary Equilibrium

Let $G(a, z, y)$ denote the stationary distribution of agents. Occupation is a function of the current state and so is not a separate coordinate of G . Define

$$\mathcal{E} = \{(a, z, y) : V_e(a, z, y) \geq V_w(a, z, y)\}, \quad (23)$$

$$\mathcal{W} = \{(a, z, y) : V_e(a, z, y) < V_w(a, z, y)\}. \quad (24)$$

Aggregate input demands and labor supply are

$$K_n^d = \int_{\mathcal{E}} K_n(a, z) dG, \quad K_u^d = \int_{\mathcal{E}} K_u(a, z) dG, \quad (25)$$

$$L^d = \int_{\mathcal{E}} L(a, z) dG, \quad L^s = \int_{\mathcal{W}} y dG. \quad (26)$$

Total deposits are $D = \int a dG$. By the balance-sheet condition (21), deposits equal the market value of the capital the intermediary holds. Combining this with the supply ratio (22) gives both stocks as functions of aggregate wealth:

$$K_n^s = \frac{D}{p_n + p_u \zeta / (\delta + \kappa)}, \quad K_u^s = \frac{\zeta}{\delta + \kappa} K_n^s. \quad (27)$$

Aggregate external finance is the value of operated capital in excess of own net worth,

$$B^{\text{ext}} = \int_{\mathcal{E}} \max \{0, p_n K_n(a, z) + p_u K_u(a, z) - a\} dG, \quad (28)$$

and its ratio to output is the leverage moment that disciplines θ . Markets clear when

$$K_n^d = K_n^s, \quad K_u^d = K_u^s, \quad L^d = L^s. \quad (29)$$

Given (27), these three conditions are equivalent to clearing the labor market, clearing the new-capital market, and requiring aggregate demand to reproduce the technological ratio $K_n^d / K_u^d = (\delta + \kappa) / \zeta$. Used-capital clearing is then automatic.

Definition. A stationary competitive equilibrium consists of prices (W, r, R_n) , the implied capital prices $p_u = p_n - (R_n - r - \delta) / \zeta$ and $R_u = r + \delta + \kappa$, value functions $\{V_e, V_w, \Omega\}$, policy functions, firm input demands, capital stocks (K_n^s, K_u^s) , and a distribution G such that:

1. given (W, R_n, R_u, p_u) , firm input demands solve (8)–(9);
2. given prices and the continuation value Ω , consumption, saving, and occupational policies solve (10)–(15);

3. the intermediary's investment policy is optimal given (r, R_n) , which by Proposition 1 requires (19)–(18) and delivers zero profit and the balance sheet (21), with the stocks evolving according to (16)–(17);
4. G is invariant under the optimal saving and occupational policies and the transition processes (5) and (7);
5. the labor, new-capital, and used-capital markets clear as in (29).

5.6. Used Capital and Financial Constraints

The capital mix. Let $\mu(a, z) \geq 0$ be the multiplier on (9). For an interior choice of both new and used capital, the firm's first-order conditions are

$$\text{MPK}_n = p_n(R_n + \mu), \quad \text{MPK}_u = p_u(R_u + \mu). \quad (30)$$

Both capital types are used in positive amounts at any finite shadow prices. Although $\gamma > 1$ makes the two types gross substitutes, the marginal product of each remains unbounded at zero: as $K_n \rightarrow 0$ with $K_u > 0$ fixed, effective capital converges to $(1 - \phi)^{1/(\gamma-1)} K_u > 0$, so $\partial K / \partial K_n = \phi^{1/\gamma} (K/K_n)^{1/\gamma} \rightarrow \infty$, and symmetrically for K_u . Both conditions in (30) therefore hold with equality at any finite shadow prices. The two capital types are treated symmetrically. The firm pays $p_j R_j$ to rent one unit of type j and bears the additional shadow cost $p_j \mu$ for the financing capacity that the unit absorbs. Neither type receives a preferential collateral coefficient; they differ only in the market value p_j at which they enter (9). Combining (30) with (4) gives the optimal capital mix $\rho \equiv K_u / K_n$:

$$\rho(\mu) = \frac{1 - \phi}{\phi} \left[\frac{p_n(R_n + \mu)}{p_u(R_u + \mu)} \right]^\gamma. \quad (31)$$

At $\mu = 0$, this becomes $\rho^{uc} \equiv \rho(0)$. Because it depends only on market prices, all unconstrained firms choose the same mix of new and used capital, regardless of their wealth or productivity. The mix varies across firms only through the multiplier μ .

Collateral productivity. How a firm splits its capital between new and used machines also determines the scale that a given amount of financing capacity can support. Define

$$\Lambda(\rho) \equiv \frac{K}{p_n K_n + p_u K_u} = \frac{\Psi(\rho)}{p_n + p_u \rho}, \quad \Psi(\rho) \equiv \left[\phi^{1/\gamma} + (1 - \phi)^{1/\gamma} \rho^{(\gamma-1)/\gamma} \right]^{\gamma/(\gamma-1)}. \quad (32)$$

The function $\Lambda(\rho)$ measures the units of effective capital obtained per dollar of financing capacity. Define $\Lambda^{uc} \equiv \Lambda(\rho^{uc})$. A firm with a binding constraint exhausts (9) and therefore operates $\Lambda(\rho(\mu))\theta a$ units of effective capital. A constrained firm therefore has two adjustment margins: its scale and its choice of new versus used capital, which determines how far a given collateral position stretches.

Proposition 2 (Used-capital tilt and collateral relief). *Suppose*

$$R_u > R_n, \quad (33)$$

or, equivalently, $\kappa > \zeta(p_n - p_u)$. A sufficient condition is $\zeta \leq \kappa$, which by the stationary aggregate-stock relation (22) can equivalently be written as

$$\frac{K_u^s}{K_n^s} \leq \frac{\kappa}{\delta + \kappa}.$$

Then:

(i) $\rho'(\mu) > 0$: the used-to-new capital ratio increases strictly and monotonically from ρ^{uc} toward

$$\bar{\rho} = \frac{1 - \phi}{\phi} \left(\frac{p_n}{p_u} \right)^\gamma,$$

the mix that minimizes the collateral value $p_n K_n + p_u K_u$ required per unit of effective capital.

(ii) Collateral productivity $\Lambda(\rho(\mu))$ is strictly increasing in μ on $[0, \infty)$ and converges to

$$\bar{\Lambda} = \left[\phi p_n^{1-\gamma} + (1 - \phi) p_u^{1-\gamma} \right]^{1/(\gamma-1)}. \quad (34)$$

Moreover, $\bar{\Lambda} > p_n^{-1}$ whenever $p_u < p_n$ and $\gamma > 1$.

The proof is in Appendix C. Proposition 2 implies that constrained firms choose a larger used share than otherwise identical unconstrained firms, and that this share rises with the tightness of the constraint. This is the model counterpart of the firm-level gradients in Section 4.⁴ The tilt does not require used capital to be cheaper to rent. Condition (33) is compatible with $p_u R_u > p_n R_n$: what matters is the financing capacity a unit absorbs per dollar of rental spending, not the rental payment itself.⁵ The relief is nevertheless bounded: because $\Lambda(\rho(\mu))$ converges to the finite limit in (34), tilting toward used capital economizes on financing capacity but cannot fully substitute for scale.⁶

Two margins. Used capital therefore affects both firm scale and occupational selection. Along the intensive margin, it allows constrained entrepreneurs to operate closer to their unconstrained scale, reducing the dispersion of marginal revenue products. Along the extensive margin, it raises the value of entrepreneurship for productive but poor potential entrants and shifts the occupational boundary (15) accordingly.

6. Quantitative Analysis

Assigned parameters. Table 6 lists every parameter with its value and source or target. Seven parameters are assigned outside the calibration (Panel A). Following the literature, I set annual

⁴Condition (33) holds at the parameters calibrated in Section 6.

⁵Eisfeldt and Rampini (2007) also predict that constrained firms use more used capital, through the timing of cash flows relative to purchase cost rather than through collateral capacity.

⁶The mechanism belongs to the market-value collateral framework of Rampini and Viswanathan (2013) and is closest to Rampini (2019), where an asset's durability determines the down payment it requires. Here durability varies within a single asset class that firms combine, so the response is a change in the mix rather than a move between asset classes.

depreciation $\delta = 0.06$, the span of control $\eta = 0.79$, and relative risk aversion $\sigma = 1.5$ as in Buera et al. (2011), the labor share within the variable-input bundle $\alpha = 2/3$ as in Midrigan and Xu (2014), and the annual persistence of worker ability $\rho_y = 0.95$ from Cagetti and De Nardi (2006). The remaining two govern the new-and-used capital aggregate (4). I set the CES weight on new capital to $\phi = 0.70$. This weight is pinned rather than estimated, because it is only weakly identified. The aggregate new-to-used ratio is a market-clearing condition that ζ delivers at any value of ϕ , so ϕ reaches the targeted moments only through the used-share gradient, and it moves that moment slowly: Appendix E raises ϕ from 0.70 to 0.75, the weight on new investment goods in the CES aggregator of Lanteri (2018), leaving the headline results unchanged. I set the elasticity of substitution to $\gamma = 2$. Edgerton (2011) estimates this elasticity from relative-price regressions derived from a CES composite of new and used equipment, and reports values between 1.7 and 2.0 for farm machinery and between 1.9 and 2.4 for construction machinery; $\gamma = 2$ lies in the overlap.

Jointly calibrated parameters. Seven parameters are calibrated jointly in the benchmark (Panel B): $\{\theta, \eta_p, \omega, \sigma_y, \beta, \kappa, \zeta\}$. Each is disciplined primarily by one moment, though all seven are selected together. The debt-to-output ratio of 0.600, computed as B^{ext}/Y from (28), disciplines the collateral multiplier θ . The employment share of the largest decile of firms, 0.632, disciplines the Pareto-tail parameter η_p of entrepreneurial productivity. The annual firm exit rate of 0.098, which I compute from the survey and which is consistent with the 9–10 percent annual death rate reported by Berkel et al. (2020), disciplines productivity persistence ω . Agents in the model are infinitely lived and firms do not die, so the moment is mapped to the rate at which entrepreneurs switch to wage work; the survey records the closure or suspension of the household’s enterprise rather than the disappearance of the household, which is what that switch is here. The worker wage Gini of 0.372, reported by Doan et al. (2023) from the Vietnam Household Living Standard Survey, disciplines the innovation dispersion σ_y of worker ability. The annual real interest rate of 0.031, the Vietnamese deposit rate net of GDP-deflator inflation in 2013 (World Bank, 2026), is matched by the discount factor: the model requires $\beta = 0.858$ to clear the deposit market at that rate. I use 2013 because the used-share gradient, the top-decile employment share and the new-to-used capital ratio are all measured in the 2013 survey round; the 2012–2016 average is 3.25 percent. The value $\beta = 0.858$ is low relative to the annual discount factors used in economies without entrepreneurs, but close to the 0.865 that Cagetti and De Nardi (2006) calibrate in a model of the same type. Productive agents earn a return on internal funds far above the deposit rate and save aggressively out of precaution, so clearing the deposit market at $r = 0.031$ requires more impatience than the deposit rate alone would suggest.

Used-capital parameters. Two moments jointly determine κ and ζ . The first is the aggregate ratio of new to used capital, which in a stationary equilibrium is

$$\frac{K_n^s}{K_u^s} = \frac{\delta + \kappa}{\zeta}, \quad (35)$$

and captures the relative *quantity* of the two capital types in the economy. The second captures the distribution of new and used capital across firms using the cross-sectional relationship between the used-capital share and firm size:

$$s_{u,i} = b_0 + b_1 \log A_i + \varepsilon_i, \quad b_1 = -0.027 \text{ (0.0061)}, \quad N = 886. \quad (36)$$

I estimate this relationship by ordinary least squares using the 2013 cross-section of firms reporting a positive used-capital share. The dependent variable $s_{u,i}$ is the used-capital share on a unit interval, so $b_1 = -0.027$ means that a one-log-point increase in assets is associated with a 2.7 percentage-point decline in the used share, or equivalently -0.027 on the 0–1 scale. This is a *between-firm* size gradient in one cross-section. It is the natural target for a stationary model and plays a different role from the within-firm zero-debt interactions in Table 5, which identify off firms whose debt status and assets change over the panel. Whereas the first moment captures the relative quantity of new and used capital, the second primarily reflects their relative price in collateral terms. If used capital is cheaper, smaller firms, which are more likely to face binding collateral constraints, rely more heavily on it, making b_1 more negative. Conversely, a smaller price advantage for used capital produces a flatter size gradient. The two moments imply $\kappa = 0.143$ and $\zeta = 0.079$.

Two readings of the survey item are possible and the choice matters for the first moment. The questionnaire asks for the percentage of a firm’s main equipment acquired new, second-hand, or self-constructed, without saying whether the percentage is of the number of machines or of their value, and it records neither a count of machines nor an asset register, so the data cannot settle it. I read the percentage as a share of equipment quantity, which is the object that (35) restricts, and aggregate it across firms weighting by each firm’s machinery at market prices, as a proxy for the equipment-quantity weights that aggregation in the model implies. Under this reading second-hand equipment is 28.0 percent of the machines in the economy but, at $p_u = 0.645$, 20.1 percent of the value of the equipment stock; the value share is not targeted. Reading the percentage as a value share instead would restrict $p_n K_n^s / (p_u K_u^s)$ rather than K_n^s / K_u^s and would imply a larger second-hand sector.

At these values, the condition of Proposition 2 is satisfied: $R_u = r + \delta + \kappa = 0.234$ exceeds $R_n = r + \delta + \zeta(p_n - p_u) = 0.119$, and the sufficient condition $\zeta \leq \kappa$ holds as well.

Comparison design. I compare two economies to isolate the role of used capital: a benchmark in which firms rent both new and used capital, and an economy without used capital. Both are calibrated to the same five common moments; the benchmark additionally matches the used-capital-share gradient and the aggregate new-to-used ratio. Removing used capital ($\zeta = 0$, $\phi = 1$) leaves the five common moments to be matched by recalibrating $\{\theta, \eta_p, \omega, \sigma_y, \beta\}$, and I solve for the general-equilibrium prices and allocations separately in each economy. Because both economies attain the same fit to the common targets, the differences between them reflect the absence of used capital. The lower value of θ is the value required to reproduce the benchmark’s debt-to-output ratio.

Table 6. Calibration: parameters, values, and targets

Parameter	Benchmark	Model without used capital	Source or target
<i>Panel A: Parameters assigned from the literature</i>			
α : Labor share	0.667	0.667	Midrigan and Xu (2014)
δ : Depreciation rate	0.060	0.060	Buera et al. (2011)
η : Span of control	0.790	0.790	Buera et al. (2011)
σ : Risk aversion	1.500	1.500	Buera et al. (2011)
ρ_y : Ability persistence	0.950	0.950	Cagetti and De Nardi (2006)
ϕ : CES weight on new capital	0.700	1.000	Assigned; see text
γ : Substitution, new vs. used	2.000	—	Edgerton (2011)
<i>Panel B: Parameters jointly calibrated to data moments</i>			
θ : Collateral multiplier	2.143	1.985	Debt-to-output ratio: 0.600
η_p : Productivity-tail parameter	5.949	5.397	Top-decile employment share: 0.632
ω : Productivity persistence	0.860	0.837	Firm exit rate: 0.098
σ_y : Ability innovation dispersion	0.245	0.240	Worker wage Gini: 0.372
β : Discount factor	0.858	0.854	Real interest rate: 0.031
ζ : New-to-used transition rate	0.079	0.000	New-to-used capital ratio: 2.570
κ : Breakage rate of used capital	0.143	—	Used-share gradient: -0.027

Notes: Most targets are computed on the Vietnamese Small and Medium Enterprise Survey (CIEM et al., 2005–2015). The two economies are calibrated separately, as described in the text.

6.1. The Mechanism of Used Capital

Used capital works through the collateral constraint. A firm that cannot meet the constraint operates below its unconstrained scale. The multiplier μ measures the shadow cost of the missing financing capacity. The left panel of Figure 1 plots this shadow cost by benchmark entrepreneur wealth decile, separately for low, mid, and high entrepreneurial productivity z . It is highest for entrepreneurs who combine high productivity with little wealth—precisely the firms for which the constraint binds most strongly—and declines with wealth but remains positive for most entrepreneurs in the stationary distribution as wealth accumulates.

The right panel maps the shadow cost into the capital mix. An unconstrained firm faces $\mu = 0$ and chooses the mix of new and used capital that minimizes its rental cost at prevailing prices. Constrained firms depart from this benchmark. A dollar of rental spending on capital type j purchases $1/(p_j R_j)$ units, each of which absorbs p_j units of financing capacity. Consequently, a dollar of rental spending on type j uses $1/R_j$ units of financing capacity. Because $R_u > R_n$, used capital absorbs less financing capacity per dollar of rental spending than new capital. The used-capital share therefore rises monotonically with μ and approaches the ceiling implied by $\bar{\rho}$ in Proposition 2.

Taken together, the two panels show that the firms facing the tightest collateral constraints are precisely those that tilt most strongly toward used equipment. This tilt is a response to financing constraints rather than a difference in production technology. By reallocating financing capacity toward used equipment, constrained firms can obtain more effective capital from a given collateral position. This mechanism drives the allocation and entry results that follow.

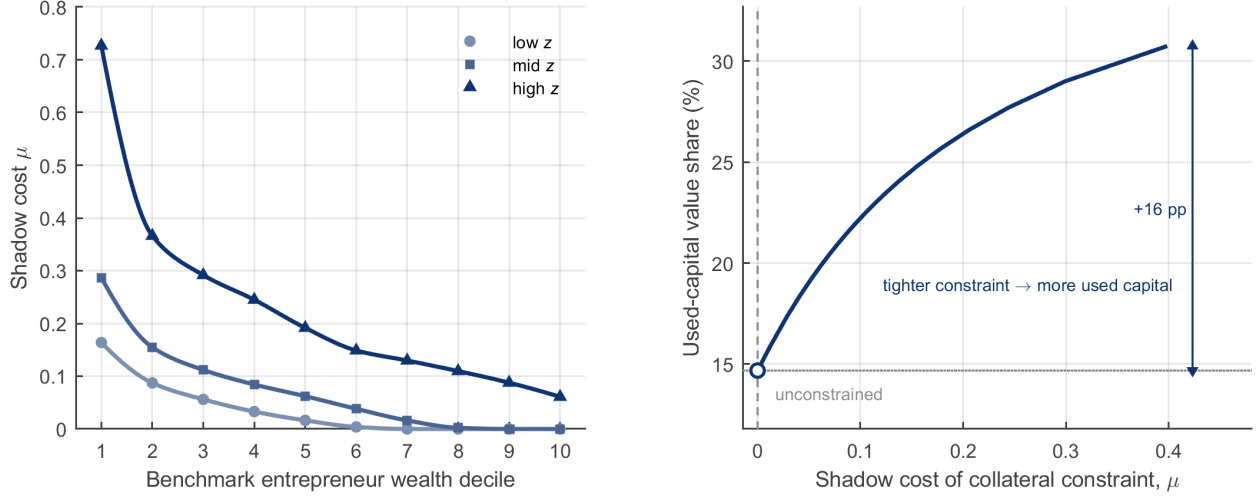


Figure 1. The mechanism of used capital. Left panel: the shadow cost of the collateral constraint, μ , by benchmark entrepreneur wealth decile, for low, mid, and high entrepreneurial productivity z (the bottom, middle, and top thirds of the productivity grid); each marker is the average μ of entrepreneurs in that wealth decile and productivity group. Right panel: the used-capital value share, $100 p_u K_u / (p_n K_n + p_u K_u)$, as a function of μ at benchmark equilibrium prices. Constrained firms are sorted by μ into 20 bins of equal size and averaged within each bin. The horizontal axis ends at the 95th percentile of μ among constrained firms, and the open circle marks the unconstrained firm at $\mu = 0$.

6.2. Used Capital and Capital Misallocation

I measure the cost of financial frictions by comparing each calibrated economy with its own frictionless counterpart. In each case I set θ large enough that the collateral constraint is slack for every operating firm. I hold all other parameters fixed and recompute equilibrium prices and the stationary distribution. No agent with strictly positive wealth is then constrained, and MRPK dispersion is numerically zero. Using a separate frictionless counterpart for each economy isolates the misallocation caused by financial frictions from differences in technology, calibration, and equilibrium prices.

Define the TFP loss from the collateral constraint as

$$\mathcal{M} = 100 \log \left(\frac{\text{TFP}^{\text{frictionless}}}{\text{TFP}^{\text{friction}}} \right), \quad (37)$$

measured in log points against the economy's own frictionless counterpart.

Relative to its own frictionless counterpart, the collateral constraint generates a TFP loss of $\mathcal{M} = 5.72$ log points in the benchmark. The same constraint generates $\mathcal{M} = 9.39$ log points in the model without used capital, recalibrated to the same targets. Second-hand capital therefore offsets 3.67 log points, or

$$\frac{9.39 - 5.72}{9.39} = 39.1\%$$

of the frictional loss that arises in its absence. This decomposes the model's misallocation into the portion the used-capital margin removes. Table 7 reports this comparison together with the

allocation statistics behind it.

Table 7. Used capital and capital misallocation

	Benchmark	No used capital	Effect of used capital
TFP loss, $\mathcal{M}$ (log points)	5.72	9.39	39.1% eliminated
Mean scale relative to target, K/K^*	0.343	0.279	23.0% higher
sd(log MRPK)	0.450	0.598	24.7% lower
Share of firms constrained	0.908	0.909	0.2% lower

Notes: The benchmark contains new and used capital, while the other model is without used capital and recalibrated to the common targets. The TFP loss $\mathcal{M}$ is defined in (37) and is computed within each economy against its own frictionless counterpart, so the two columns are not measured against a common benchmark. K/K^* compares each firm’s new capital to the amount it would install if the collateral constraint did not bind at the economy’s prices; the table reports the average across firms (very high ratios are capped at 1.2 so a few outliers do not dominate). A value of one means firms operate at that unconstrained scale on average; lower values mean they fall short. The sd(log MRPK) row measures how much marginal products differ across firms: a higher number means more dispersion and more misallocation. Percentages in the last column compare the benchmark with the model without used capital and are computed from the unrounded values. The share of firms constrained counts entrepreneurs for whom the collateral constraint binds at the margin, $\mu > 0$, rather than firms refused credit. A firm that finances itself because the constraint would bind at any larger scale is constrained here and appears in the survey as a non-applicant.

Second-hand capital reallocates capital toward financially constrained, high-return firms. Access to used capital allows constrained entrepreneurs to operate closer to their unconstrained scale, as Figure 2 shows. The effect is largest among low-wealth entrepreneurs. Second-hand capital therefore directs financing capacity toward firms with the highest marginal returns to capital.

The single-vintage economy is separately calibrated. The headline estimate therefore also incorporates changes in the fitted productivity distribution. Holding that distribution fixed instead gives 19.5 percent. Appendix D reports all three comparison designs. No margin changes sign across them.

6.3. Used Capital and Entry by Productive Poor Agents

The second main result concerns occupational selection. I define “talented” agents as those in the top decile of entrepreneurial productivity z . “Poor” agents are those with wealth below the benchmark median. Access to used capital raises entrepreneurship among poor talented agents from 34.3 to 42.0 percent, an increase of 7.7 percentage points. By contrast, entrepreneurship among rich talented agents is nearly unchanged. Used capital therefore reduces the dependence of entrepreneurship on initial wealth rather than increasing entry uniformly. Table 8 reports these entry rates together with their frictionless counterparts.

The mechanism operates through the occupational boundary. Used capital absorbs less financing capacity. Productive agents with little wealth can therefore operate at a larger scale. This raises profits $\pi(a, z)$ and the value of entrepreneurship $V_e(a, z, y)$, enabling some of them to enter. Rich talented agents, those with wealth above the benchmark median, may still face a binding constraint at the margin, but they are wealthy enough to enter even at a constrained scale. Their occupational

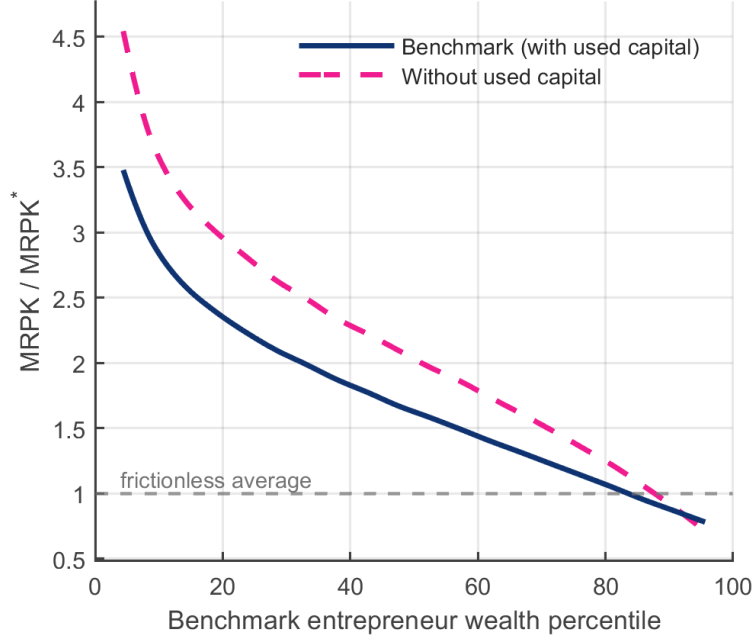


Figure 2. Used capital and the allocation of capital. The marginal revenue product of capital (MRPK) divided by the average MRPK in that economy’s frictionless counterpart (MRPK*), plotted by benchmark entrepreneur wealth percentile. Entrepreneurs are grouped into 20 bins between the 2nd and 98th percentiles of benchmark entrepreneur wealth, and the same wealth cutoffs are used for the model without used capital. Within each bin, the median MRPK at each productivity level is averaged over the economy’s productivity distribution of entrepreneurs, so the profile reflects financial constraints rather than sorting by ability; lines interpolate between bin midpoints. Solid line: benchmark with used capital; dashed line: model without used capital. MRPK* is an economy-wide average in the frictionless economy, not the MRPK of an unconstrained firm at the same prices; a ratio near one therefore does not mean the collateral constraint is slack ($\mu = 0$), and the ratio falls below one for the wealthiest entrepreneurs.

Table 8. Used capital and entry by productive poor agents

	Benchmark	No used capital	Effect of used capital
Poor talented entrepreneurship rate	0.420	0.343	+7.7 p.p.
Rich talented entrepreneurship rate	0.827	0.833	−0.6 p.p.
Poor talented rate, frictionless	0.714	0.682	—
Rich talented rate, frictionless	0.801	0.744	—
Entry shortfall, $\mathcal{S}$ (log points)	53.2	68.7	22.6% repaired

Notes: “Talented” denotes the top productivity decile and “poor” denotes wealth below the benchmark population median. These cutoffs are constructed once in the benchmark and applied in levels to both economies. Each economy is compared with its own frictionless counterpart, so the frictionless rows differ across columns. The entry shortfall $\mathcal{S}$ in (38) is defined for poor talented agents.

choices therefore change little (0.827 with used capital, 0.833 without).

To account for poor talented agents who prefer wage work even without financial frictions, I define the collateral-induced entry shortfall as

$$\mathcal{S} = 100 \log \left(\frac{e^{\text{frictionless}}}{e^{\text{friction}}} \right), \quad (38)$$

where e is their entrepreneurship rate. The shortfall is $\mathcal{S} = 53.2$ log points in the benchmark and $\mathcal{S} = 68.7$ log points in the model without used capital. Thus, used capital repairs

$$1 - \frac{53.2}{68.7} = 22.6\%$$

of the entry distortion caused by the collateral constraint.

For rich talented agents, the shortfall is negative in both economies. Entry under the constraint (0.827 in the benchmark) exceeds entry in its own frictionless counterpart (0.801), implying a shortfall of -3.2 log points. The corresponding figure without used capital is -11.3 log points. This is a general-equilibrium effect rather than a relaxation of the constraint. Removing the collateral constraint raises labor demand and hence the wage. This raises the value of wage work and draws some marginal high-productivity agents out of entrepreneurship. The agents affected are those who enter even under frictions: the constraint does not block their entry, although it may still limit their scale ($\mu > 0$).

7. Conclusion

This paper begins from a discrepancy between standard models and the data. In models with homogeneous capital, a financially constrained firm can respond to limited credit only by operating at a smaller scale. In the data, however, young, small, and credit-excluded firms also change the mix of their capital. They rely more heavily on second-hand equipment. This paper studies whether that additional adjustment margin changes the allocation and occupational consequences of financial frictions.

The mechanism follows from the way collateral constraints are defined. The constraint limits the market value of the capital a firm operates. Used capital is more expensive to operate, because it depreciates faster. But its market price is lower. A dollar spent renting used equipment therefore ties up less collateral than a dollar spent renting new. A productive entrepreneur with little wealth can use second-hand equipment to operate more capital than new equipment alone would allow. As the entrepreneur accumulates wealth, the constraint becomes less severe. The advantage then declines, and the firm shifts toward the unconstrained mix. This mechanism accounts for the concentration of used equipment among young, asset-poor, and credit-excluded firms in the Vietnamese data.

Quantitatively, access to second-hand capital reduces the misallocation that the collateral constraint generates by 39 percent. The comparison is with a single-vintage economy in which firms rent only new capital. It is calibrated on its own to the same five empirical targets; because the vintage margin is absent, its calibrated parameters differ from the benchmark's. Used equipment allows productive firms with little wealth to operate closer to their unconstrained scale, and it reduces the dispersion of marginal products across firms. The same mechanism affects occupational choice. Without access to used capital, productive but poor agents are less likely to become entrepreneurs. Entry among productive and wealthy agents changes little.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) in order to improve the language and readability of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Data availability

The Viet Nam SME Survey (2005–2015), collected by CIEM, ILSSA, the Development Economics Research Group at the University of Copenhagen, and UNU-WIDER, is distributed by UNU-WIDER at <https://www.wider.unu.edu/database/viet-nam-sme-database>. The enterprise data files are supplied free of charge on completion of UNU-WIDER’s data request form; the author had no access privileges beyond those available to any researcher. Because UNU-WIDER distributes the files under its own request procedure, they are not included in the replication package. The code that constructs the empirical moments, solves and calibrates the model, and produces every table and figure in the paper is publicly available at <https://doi.org/10.5281/zenodo.22947884> (Rahimfallah, 2026). The solution method follows the computational approaches of Midrigan and Xu (2014) and Tan (2026), and function approximation uses the CompEcon toolbox of Miranda and Fackler (2002), which is not redistributed with the package.

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Appendix A. Credit Constraints and Used Capital

This appendix uses the credit module of the Vietnam SME Survey to measure credit constraints directly and relates them to firms’ equipment mix. It documents three facts that correspond to the mechanism of Section 5.6.

1. *Constrained firms are young and have few assets for their size.* Firms aged 11 years or more are 6.8 points less likely to report a credit constraint than firms aged five or less, on a mean of 17 percent. At a given number of workers, firms with more assets are less likely to be constrained.
2. *Constrained firms hold more used and less new equipment.* Comparing each firm with itself, a constrained firm holds 1.7 points more of its main equipment second-hand. Among firms that borrow and are constrained, the used share rises by 2.7 points and the new share falls by 2.3. When the reported problem is a lack of collateral, the used share rises by 6.0 points and the new share falls by 6.6.
3. *Young firms move away from used capital as they grow.* Firms first observed at five years or less build up assets faster than older firms, by 8 percent every two years. Their used share falls faster, by 1.3 points every two years and about 6 points over a decade.

The evidence is descriptive.

A.1. Sample and Measures

The sample is that of Panel B of Table 1: firms that use machinery and whose new, second-hand and self-constructed percentages add up to 100, 14,691 firm-rounds from 4,905 firms. Of these firms, 3,495 are observed in at least two rounds and 1,486 change credit status at least once. That within-firm variation provides the variation used for the fixed-effects estimates below.

The main outcome is the used share of Section 3.2. I also report the new and self-built shares, which show what the used share replaces. *Used relative to new*, $100 \times u/(u + n)$, is a monotone transform of the used-to-new ratio $\rho = K_u/K_n$ in Proposition 2, so it is the closest data counterpart of the model’s vintage choice. A further outcome captures the extensive margin: whether any of the main equipment was acquired second-hand. The item describes the stock of main equipment, not the flow of purchases in the last two years, so it moves slowly and changes in credit status are matched to changes in a stock.

I measure constraints directly from the credit module rather than inferring them from debt. Every round asks whether the firm has applied for bank loans or other formal credit since the last survey; since 2002 in the 2005 round. Applicants are asked whether they had problems getting the loan and, if so, the main reason; from 2007 the survey also counts denied applications. Non-applicants are asked why they did not apply. The answer codes are identical in all six rounds, and the wording differs only in the reference period (Table 10).

A firm is *credit constrained* in a round if either condition holds:

- it applied and reported problems obtaining the loan, or had an application denied;
- it did not apply because its collateral was inadequate or the process too difficult.

This measure follows the direct survey-based approach used in the credit-constraint literature, including Rand (2007) for the Vietnam SME surveys and Kuntchev et al. (2013) for the World Bank Enterprise Surveys, while adapting the classification to the response categories available in these surveys. Two groups are counted as unconstrained because their responses indicate either the price of credit or an absence of demand for formal borrowing rather than the collateral- or procedure-related restrictions captured by the constrained category:

- firms that did not apply because the interest rate was too high or they were already heavily indebted;
- firms that did not need a loan or did not want debt.

Table 9 describes credit status. On average 17.2 percent of firm-rounds are constrained, falling from 22.1 percent in 2005 to 10.7 percent in 2015. In the pooled data, constrained firms hold more used equipment (26.9 against 24.0 percent) and less new equipment (69.7 against 72.7). They are also larger, younger, and far more likely to have debt (72.1 against 49.7 percent). The measure is therefore not a relabeling of zero debt: most non-applicants without debt say they did not need a loan.

Table 9. Credit status and the equipment mix

<i>Panel A. Credit status by survey round</i>							
	2005	2007	2009	2011	2013	2015	All
Applied for formal credit (%)	40.3	38.1	38.8	30.6	26.9	25.5	33.4
Problems or denial, among applicants (%)	19.6	21.0	21.9	27.3	24.5	15.1	21.5
Discouraged, among non-applicants (%)	23.7	17.1	17.3	11.6	13.0	9.2	15.0
Credit constrained (%)	22.1	18.6	19.1	16.5	16.1	10.7	17.2
Has debt (%)	61.1	56.5	60.5	51.8	52.4	39.1	53.6
<i>Panel B. Means by credit status, all rounds</i>							
	Constrained		Unconstrained		Difference		
Used share of main equipment (%)	26.9		24.0		+2.9		
New share (%)	69.7		72.7		−3.0		
Self-built share (%)	3.4		3.3		+0.1		
Used relative to new, $u/(u + n)$ (%)	28.1		25.2		+2.9		
Any used equipment (%)	41.3		34.8		+6.5		
Full-time workers	19.1		15.5		+3.5		
Firm age (years)	13.1		15.0		−1.8		
Has debt (%)	72.1		49.7		+22.3		
Firm-rounds	2,516		12,128				

Notes: Firms that use machinery and whose new, second-hand and self-constructed percentages add up to 100, the sample of Panel B of Table 1. Panel A: shares of firm-rounds in percent. “Problems or denial” is computed among firms that applied for formal credit since the previous round. “Discouraged” is computed among non-applicants whose reason is inadequate collateral or a process that was too difficult. *Credit constrained* is either of the two, as a share of all firm-rounds. *Has debt*: positive outstanding debt at the end of the previous year. Panel B pools all rounds; differences are unconditional. Firm age is measured from the establishment year the firm first reports.

Table 10. Credit and equipment items by survey round

Item (2015 wording)	2005	2007	2009	2011	2013	2015
“What percentage of your main equipment/machinery did you get?” New / used or second hand / self-constructed	q34a–c	q30a–c	q30aa–ac	q30aa–ac	q30aa–ac	q31aa–ac
“Has your firm applied for bank loans or other formal credit since [the last survey]?”	q109 ^a	q99	q99	q80	q80	q70
“Did your firm experience any problems getting the loan?”	q110a	q101	q100	q81	q81	q71
“If yes, why?” (1 = lack of collateral)	q110b	q101a	q100a	q81a	q81a	q71a
“How many formal loan applications have been denied since [the last survey]?”	—	q104a–b	q103a–b	q84a–b	q84a–b	q74a–b
“Why has the firm not applied for formal loans since [the last survey]?”	q117	q100	q106	q87	q87	q77
Total outstanding debt, end-year	Eq1u	Eq1v	EAq1v	EAq1v	EAq1v	EAq1i
Total assets, end-year ^b	Eq1p	Eq1q	EAq1q	EAq1q	EAq1q	EAq1h
Full-time workers, end-year	Eq1v	Eq1x	EAq1x	EAq1x	EAq1x	EAq1k

Notes: Question numbers in the round files, with the 2015 wording. Answer codes are identical across rounds. For problems, code 1 is *lack of collateral*. Reasons for not applying are coded 1 inadequate collateral, 2 do not want to incur debt, 3 process too difficult, 4 did not need one, 5 interest rate too high, 6 already heavily indebted, 7 other. ^a In 2005 the reference period is “since 2002”. ^b Physical assets at accounting value in 2005 and 2007, at market prices in 2009–2013, and total assets in 2015; round effects absorb the level shifts. In 2011 and 2013 the variable labels of the denial items read “applications”, but the questionnaire asks for applications denied.

A.2. Specification

I estimate

$$y_{it} = \chi \text{Constrained}_{it} + \mathbf{\Gamma}' \mathbf{Z}_{it} + \lambda_t + \nu_i + \varepsilon_{it}, \quad (\text{A1})$$

where y_{it} is an equipment-mix outcome of firm i in round t , and λ_t and ν_i are round and firm fixed effects, as in (1). The controls $\mathbf{Z}_{it}$ are:

- log employment (full-time workers), which measures size continuously;
- log assets, centered at the sample mean;
- indicators for firm age of 6–10 and of 11 or more years, with firms aged five years or less as the omitted group.

Firm age is computed from the establishment year that a firm first reports; Table 2 uses its most frequent answer instead. Either way the year is fixed within a firm, so every firm ages exactly two years between rounds and a linear age term is collinear with the firm and round effects. Age therefore enters in bands, which are identified by the 1,673 firms that cross an age threshold during the panel. Table 15 shows that the results do not depend on how size and age are controlled for.

Proposition 2 gives the corresponding model prediction. The used-to-new ratio ρ rises with the shadow cost μ of the collateral constraint. A constrained firm ($\mu > 0$) therefore chooses a larger

used share than an otherwise identical unconstrained firm ($\mu = 0$). The empirical counterparts are $\chi > 0$ for the used share and used relative to new, and $\chi < 0$ for the new share.

The firm effects absorb characteristics of a firm that do not change over the decade, such as its product, technology, location, management and the permanent component of its access to credit. The round effects absorb aggregate conditions and changes in the questionnaire, including the switch from book to market values of assets in 2009. The coefficient χ therefore comes from comparing the same firm across rounds in which its reported credit status differs, conditional on the included controls and common round effects.

Standard errors are clustered by firm. Firm effects are nested within the clusters, so they are not counted in the degrees-of-freedom correction, as in `xtreg`, `fe` and `reghdfe` (Cameron and Miller, 2015). Firms observed in only one round are dropped because they carry no within-firm information.

Facts 1 and 3 use two further designs, described where they are used. Fact 1 asks which firms report constraints, so it compares firms with each other within the same province, round and sector. Fact 3 follows firms that are first observed young (Appendix A.5).

The estimates are descriptive. Self-reported credit status may capture changes in firms' circumstances or plans that also affect their equipment mix, so the within-firm estimates should not be interpreted as causal effects of credit constraints. The timing tests in Appendix A.4 examine whether changes in the equipment mix systematically precede changes in reported credit status.

A.3. Fact 1: Constrained Firms Are Young and Have Few Assets for Their Size

In the model, the shadow cost of the collateral constraint is highest for productive entrepreneurs with little wealth relative to the scale they want to operate (Figure 1). New firms start with little wealth, so the model predicts that constraints are most likely to bind for young firms and, at a given scale, for firms with few assets. Table 11 asks whether firms that report constraints display these patterns. All columns compare firms in the same province, round and sector.

- *Age.* Firms aged 6–10 are 3.4 points less likely to be constrained than firms aged five or less, and firms aged 11 or more are 6.8 points less likely (column 1), on a mean of 17.2 percent.
- *Assets.* At a given number of workers, one log point more assets is associated with a 1.25-point lower probability of reporting a constraint (s.e. 0.28, column 2). Employment measures the scale at which a firm operates; assets measure what it can pledge.
- *Scale relative to assets.* At a given level of assets, firms with more workers are more likely to report a constraint (+3.2 points per log point of employment). Together with the asset gradient, this pattern is consistent with constraints being more common where the scale of the business is large relative to what it can pledge. In the model, the multiplier on the constraint is highest where desired capital far exceeds wealth.

Column (3) looks at firms that did not apply for formal credit. Being put off by collateral or procedures falls with age (−7.75 points for firms aged 11 or more) and with assets (−1.41 per log point).

Table 11. Who is constrained

	Credit constrained		Put off
	Age only (1)	Age and size (2)	applying (3)
Firm age 6–10 years	−3.41*** (1.15)	−3.33*** (1.14)	−4.39*** (1.38)
Firm age 11 years or more	−6.79*** (1.09)	−6.26*** (1.09)	−7.75*** (1.31)
Log assets		−1.25*** (0.28)	−1.41*** (0.31)
Log employment		3.22*** (0.42)	4.02*** (0.49)
Mean of dependent variable	17.2	17.2	15.0
Observations	14,587	14,562	9,674
Firms	4,889	4,887	4,017
Sample	All firms	All firms	Non-applicants
Province × round and sector effects	Yes	Yes	Yes

Notes: Columns (1) and (2): 100× the credit-constraint indicator. Column (3): 100× an indicator for not applying because collateral was inadequate or the process too difficult, among firm-rounds that did not apply for formal credit. OLS with province-by-round and sector effects, so each column compares firms in the same province, round and sector. Age bands are relative to firms aged five years or less; log assets is centered. Standard errors clustered by firm in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.4. Fact 2: Constrained Firms Hold More Used and Less New Equipment

Table 12 reports the within-firm estimates for the used share. In rounds when a firm reports being constrained, the share of its main equipment acquired second-hand is 1.7 percentage points higher than in rounds when it does not (columns 1 and 2). That is about 7 percent of the sample mean of 24.0. Adding the controls leaves the coefficient essentially unchanged.

The association is also robust to more flexible controls for local conditions. Replacing the round effects with sector-by-round effects (column 3) gives 1.56, and province-by-round effects (column 4) give 1.51, both significant at the 10 percent level. Column (5) splits the measure:

- firms that applied and reported problems or a denial hold 2.81 points more used equipment (s.e. 1.24);
- firms discouraged from applying by collateral or procedures hold 0.92 points more (s.e. 1.09).

Used instead of new. Table 13 shows which part of the equipment mix moves. In Panel A, the used share rises by 1.7 points, while the new share falls by 1.2 and the self-built share by 0.6, and the share of firms holding any used equipment rises by 2.8 points on a mean of 35.3. The estimates therefore indicate that the higher used share is accompanied primarily by a lower new-equipment share.

Panel B examines borrowing status. In the model, a constrained entrepreneur borrows up to its limit, so the closest data counterpart is a firm that has debt and still reports being constrained. For

Table 12. Credit constraints and the used-capital share: within-firm estimates

<i>Dependent variable:</i> used share of main equipment (percentage points)					
	(1)	(2)	(3)	(4)	(5)
Credit constrained	1.70** (0.86)	1.72** (0.86)	1.56* (0.87)	1.51* (0.87)	
Applied: problems or denial					2.81** (1.24)
Did not apply: collateral or procedures					0.92 (1.09)
Log employment		−0.82 (0.75)	−0.86 (0.76)	−0.80 (0.75)	−0.84 (0.75)
Log assets		−0.71* (0.40)	−0.80* (0.41)	−1.17*** (0.41)	−0.72* (0.40)
Firm age 6–10 years		−1.11 (1.37)	−0.48 (1.39)	−0.25 (1.37)	−1.13 (1.37)
Firm age 11 years or more		−1.88 (1.86)	−0.92 (1.90)	−0.27 (1.86)	−1.90 (1.86)
Mean of dependent variable	24.0	24.0	24.1	24.0	24.0
Observations	13,231	13,197	13,160	13,197	13,197
Firms	3,490	3,487	3,485	3,487	3,487
Within-firm R^2	0.036	0.036	0.055	0.071	0.036
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Round fixed effects	Yes	Yes			Yes
Sector × round fixed effects			Yes		
Province × round fixed effects				Yes	

Notes: OLS with fixed effects. *Credit constrained:* the firm applied for formal credit since the previous round and reported problems obtaining it or had an application denied, or it did not apply because its collateral was inadequate or the process too difficult. Column (5) enters the two parts separately. Controls: log employment, centered log assets, and indicators for firm age of 6–10 and 11 or more years (omitted: five years or less). Within-firm R^2 includes the round, sector-by-round or province-by-round effects. Round effects are not shown. Standard errors clustered by firm in parentheses. Firm effects are nested in the clusters and are not counted in the degrees-of-freedom correction; counting them, as **areg** does, multiplies every standard error by about 1.17. Firms observed in one round only are dropped. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

these firms, the used share rises by 2.75 points (s.e. 1.01) and the new share falls by 2.32 (s.e. 1.07); used relative to new rises by 2.43 points (s.e. 1.04); and the share holding any used equipment rises by 3.8 points. Constrained firms without debt, which are less directly comparable to the model’s constrained entrepreneurs, show no corresponding increase in the used share (-0.61 , s.e. 1.61). The difference between the two groups is 3.4 points (s.e. 1.9, $p = 0.08$).

Panel C isolates reported collateral problems. Among loan applicants, a firm whose reported problem was a lack of collateral holds 6.0 points more used equipment (s.e. 2.6) and 6.6 points less new equipment (s.e. 2.8) than an applicant without problems. Its used relative to new is also 6.4 points higher (s.e. 2.7). Problems for other reasons, such as paperwork, regulations or the bank’s assessment of the business plan, are associated with a 1.4-point increase in the used share (s.e. 1.8), which is not statistically significant. The collateral margin is therefore the empirical margin most closely aligned with the model’s borrowing constraint.

Timing. Table 14 looks at changes between consecutive rounds. The point estimates have the predicted signs: becoming constrained is associated with an increase in the used share, while leaving the constrained state is associated with a decrease. Entering the constrained state raises the probability of holding any used equipment by 3.6 points (s.e. 1.8).

Columns (3) and (4) use pairs of rounds that are followed by another round and add a placebo: the change in status between the current and the next round. If firms were systematically shifting toward used equipment before reporting a constraint, the coefficient on future entry into the constrained state would be positive. Instead, it is -0.8 (s.e. 1.7), while in this sample becoming constrained is associated with a 5.1-point increase in the used share (s.e. 2.5). There is therefore no evidence in this specification that the shift toward used equipment systematically precedes the change in reported credit status.

In levels, adding the previous round’s status to (A1) leaves the coefficient on current status at 2.67 (s.e. 1.06) and gives 0.62 (s.e. 1.02) for lagged status. Thus, in this specification, the equipment mix is more strongly associated with current than lagged credit status. This pattern is consistent with the questionnaire’s timing: the credit items refer to experiences since the previous interview, while the equipment measure records the firm’s current stock of main equipment.

Robustness. Table 15 varies the sample, the measure, the controls, the fixed effects and the clustering. The coefficient on the used share stays between 1.46 and 2.34 in every specification. Dropping 2015, when constraint rates are lowest, gives 2.05; restricting to 2007–2015, when denials are recorded, gives 2.34; keeping firms seen in at least three rounds gives 2.05; dropping firms that changed owner gives 1.66; and using only reported problems, whose wording is identical in all rounds, gives 1.74. The estimates are also similar under alternative controls. Using the two size classes instead of log employment gives 1.71, dropping the asset control gives 1.72, dropping the age bands gives 1.73, and using log age instead of the bands gives 1.74. Clustering by province and sector instead of by firm raises the standard error to 1.04. With sector-by-round and province-by-round effects together the estimate is 1.46 (s.e. 0.88, $p = 0.10$). The estimated probability of holding any

Table 13. Used instead of new: the equipment mix of constrained firms

	Used share (1)	New share (2)	Self-built share (3)	Used relative to new (4)	Any used (5)
<i>Panel A. All firms</i>					
Credit constrained	1.72** (0.86)	−1.16 (0.91)	−0.57 (0.40)	1.32 (0.88)	2.83** (1.13)
Observations	13,197	13,197	13,197	12,981	13,197
<i>Panel B. By borrowing status</i>					
Constrained, has debt	2.75*** (1.01)	−2.32** (1.07)	−0.43 (0.46)	2.43** (1.04)	3.80*** (1.36)
Constrained, no debt	−0.61 (1.61)	1.30 (1.69)	−0.68 (0.74)	−1.18 (1.65)	0.20 (1.97)
Has debt	−0.16 (0.77)	0.73 (0.79)	−0.56* (0.32)	−0.32 (0.79)	0.85 (0.98)
Difference: has debt − no debt	3.36* (1.89)	−3.62* (1.98)	0.26 (0.86)	3.61* (1.94)	3.60 (2.38)
Observations	13,197	13,197	13,197	12,981	13,197
<i>Panel C. Loan applicants, by reason for the problem</i>					
Problems: lack of collateral	5.95** (2.63)	−6.60** (2.84)	0.65 (1.05)	6.39** (2.73)	8.15** (3.69)
Problems: other reasons	1.42 (1.76)	−1.98 (1.88)	0.56 (0.76)	1.88 (1.83)	2.43 (2.51)
Observations	3,714	3,714	3,714	3,655	3,714
Mean of dependent variable, Panel A	24.0	72.5	3.4	25.2	35.3

Notes: Each panel and column is a separate regression with firm and round fixed effects and the controls of Table 12. Columns (1)–(3) are the percentages of main equipment acquired used, new and self-constructed, which sum to 100. Column (4) is $100 \times \text{used} / (\text{used} + \text{new})$, the counterpart of the used-to-new ratio in Proposition 2, defined when $\text{used} + \text{new} > 0$. Column (5) is $100 \times$ an indicator for any main equipment acquired used. Panel B interacts the constraint indicator with positive outstanding debt at the end of the previous year; the difference row is the difference between the two interaction coefficients. Panel C keeps firm-rounds in which the firm applied for formal credit (1,265 firms); the omitted group is applicants without problems. Standard errors clustered by firm in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 14. Timing: switching into and out of the constrained state

	All pairs of rounds		Pairs with a next round	
	Δ Used	Δ Any	Δ Used	Δ Any
	share (1)	used (2)	share (3)	used (4)
Became constrained (0 to 1)	1.87 (1.35)	3.61** (1.80)	5.11** (2.48)	7.77** (3.33)
Stopped being constrained (1 to 0)	-1.37 (1.18)	-2.37 (1.59)	-1.49 (1.58)	-3.15 (2.13)
Becomes constrained in the next round			-0.77 (1.69)	-2.47 (2.31)
Stops being constrained in the next round			-2.96 (2.27)	-4.46 (3.01)
Observations (pairs of rounds)	9,524	9,524	6,027	6,027
Firms	3,459	3,459	2,437	2,437
Round fixed effects	Yes	Yes	Yes	Yes
Changes in the controls	Yes	Yes	Yes	Yes

Notes: First differences between consecutive rounds, two years apart. *Became constrained:* unconstrained in the previous round and constrained in the current one; *stopped being constrained:* the reverse. The omitted group is firms whose status does not change. Columns (3) and (4) keep pairs followed by a further round and add the change in status between the current and the next round as a placebo. All columns include round effects and changes in log employment, log assets and the age bands. Standard errors clustered by firm in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

used equipment is positive in every specification, ranging from 2.6 to 3.4 points.

Table 15. Robustness of the within-firm estimate

	Used share (1)	Any used (2)	Observations / firms (3)
(a) Baseline	1.72** (0.86)	2.83** (1.13)	13,197 / 3,487
(b) Excluding 2015	2.05** (0.97)	3.10** (1.26)	10,806 / 3,116
(c) Rounds 2007–2015 (denials recorded)	2.34** (0.93)	3.44*** (1.25)	10,758 / 3,080
(d) Firms observed in three or more rounds	2.05** (0.90)	2.97** (1.19)	11,225 / 2,501
(e) Firms that did not change owner	1.66* (0.86)	2.83** (1.14)	12,835 / 3,438
(f) Applicants: problems only, same wording every round	1.74** (0.86)	2.88** (1.13)	13,195 / 3,487
(g) Size classes instead of log employment	1.71** (0.86)	2.82** (1.13)	13,197 / 3,487
(h) No asset control	1.72** (0.86)	2.82** (1.13)	13,199 / 3,487
(i) No age controls	1.73** (0.86)	2.83** (1.13)	13,206 / 3,488
(j) Log of age instead of age bands	1.74** (0.86)	2.84** (1.13)	13,197 / 3,487
(k) Sector $\times$ round and province $\times$ round effects	1.46* (0.88)	2.58** (1.15)	13,160 / 3,485
(l) Clustered by province $\times$ sector	1.77* (1.04)	2.88** (1.28)	13,160 / 3,485

Notes: Each cell is the coefficient on the constraint indicator from a separate regression, specified as in Table 12, column (2), unless the row states otherwise. Rows (b)–(e) change the sample. Row (f) uses reported problems only (denials are not asked in 2005). Rows (g)–(j) change the controls: (g) uses indicators for 10–49 and 50 or more workers instead of log employment, (h) drops log assets, (i) drops the age bands, and (j) uses $\log(1 + \text{age})$ instead of the bands. Row (k) adds sector-by-round and province-by-round effects together. Row (l) clusters by province $\times$ sector (194 clusters).

Column (3) refers to column (1). Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$,

* $p < 0.10$.

A.5. Fact 3: Young Firms Move Away from Used Capital as They Grow

In the model, a firm that starts poor saves and accumulates wealth. As its wealth grows, the shadow cost μ of the borrowing constraint falls and the firm shifts its capital mix from used toward new. The survey follows the same firms for up to ten years, allowing me to examine whether their equipment mix evolves in this direction. I take the 1,509 firms first observed at age five or less and compare their subsequent paths with those of older firms observed in the same rounds:

$$y_{it} = \xi (\text{Young}_i \times \tau_{it}) + \lambda_t + \nu_i + \varepsilon_{it}, \quad (\text{A2})$$

where Young_i marks firms first observed at age five or less and τ_{it} counts the rounds since firm i was first observed. The firm effects absorb each firm's initial level, and the round effects absorb changes common to all firms. A common age trend cannot be separated from the round effects, but the differential path of young relative to older firms can. Thus, ξ measures how much faster a young firm's outcome changes, every two years, relative to the change for older firms.

Table 16, Panel A, shows two main patterns. Young firms build up assets faster: 8.0 log points more every two years (s.e. 1.7), while employment grows 1.8 log points faster. At the same time, their used share falls 1.3 points faster every two years (s.e. 0.53); used relative to new falls by 1.2 points and the probability of holding any used equipment falls by 1.4 points.

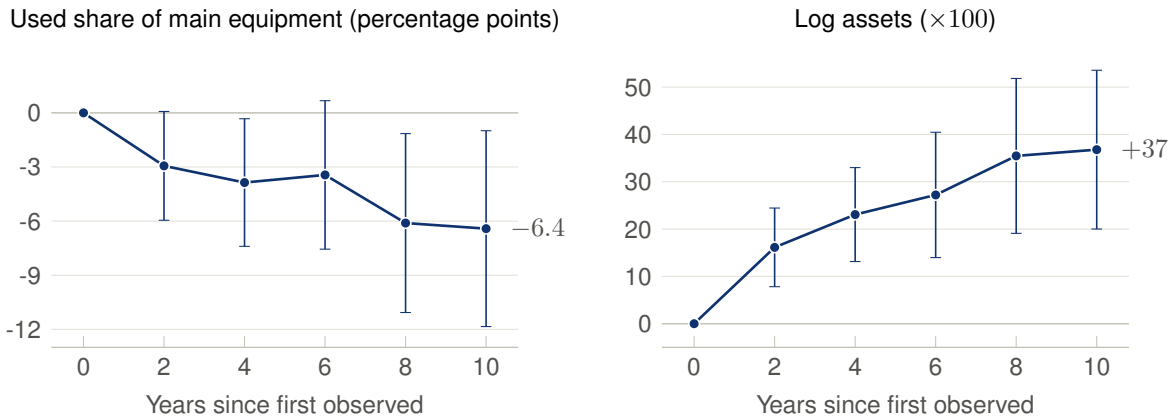
Figure 3 traces these differential paths. Relative to older firms and to the difference at the first observed round, young firms' used share is 2.9 points lower after two years and 6.4 points lower after ten. Their assets are 16 log points higher after two years and 37 after ten. In the raw data, firms first observed young start with a used share of 34.5 percent, compared with 27.8 percent for older firms, and this gap narrows over time.

Panel B checks the robustness of the differential trend. The used-share coefficient remains negative when young is defined as age three or less or seven or less (-1.66 , -1.42), with province-by-round effects (-0.96 , s.e. 0.53), without 2015 (-1.48), and among firms first observed in 2005 (-1.30). It also remains negative when size controls and reported credit status are included (-1.21), while the coefficient on the constraint indicator remains positive (1.74). These patterns are consistent with two distinct empirical relationships: firms first observed young move away from used capital as they grow, while within a firm, reported credit constraints are associated with a higher used share.

Table 16. Moving away from used capital: firms first observed young

<i>Panel A. Outcomes</i>						
	Used share (1)	Used rel. to new (2)	Any used (3)	Log assets (4)	Log empl. (5)	
Young when first observed $\times$ rounds since first observed	-1.28** (0.53)	-1.20** (0.54)	-1.40** (0.68)	7.97*** (1.66)	1.82** (0.85)	
Mean of dependent variable	24.1	25.3	35.4	—	—	
Observations	13,279	13,062	13,279	13,277	13,256	
Firms	3,494	3,468	3,494	3,494	3,492	
<i>Panel B. Used share, alternative specifications</i>						
	Young ≤ 3 yrs (1)	Young ≤ 7 yrs (2)	Prov. $\times$ round (3)	Excl. 2015 (4)	First seen 2005 (5)	Size and credit (6)
Young when first observed $\times$ rounds since first observed	-1.66** (0.73)	-1.42*** (0.47)	-0.96* (0.53)	-1.48** (0.67)	-1.30** (0.60)	-1.21** (0.53)
Credit constrained						1.74** (0.86)
Observations	13,279	13,279	13,279	10,882	8,873	13,204
Firms	3,494	3,494	3,494	3,124	2,032	3,487

Notes: Firm and round fixed effects. *Young when first observed:* first observed at age five years or less (1,509 firms). *Rounds since first observed:* survey rounds since the firm was first observed (0 to 5), so each unit is two years. In Panel A, columns (4) and (5) are $100 \times \log$ assets and $100 \times \log$ employment. In Panel B the dependent variable is the used share: (1) and (2) change the age cut-off, (3) replaces the round effects with province-by-round effects, (4) drops 2015, (5) keeps firms first observed in 2005, and (6) adds log employment, log assets and the credit-constraint indicator. Standard errors clustered by firm in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

**Figure 3.** Young firms over time: used share and assets

Notes: Coefficients on firms first observed young interacted with indicators for the years since the firm was first observed, from regressions with firm and round fixed effects. Older firms are the comparison group, and the reference point is the round in which a firm is first observed. Bars are 95 percent confidence intervals; standard errors clustered by firm. Young firms observed 2, 4, 6, 8 and 10 years after first observation: 1,044, 754, 500, 315 and 244.

Appendix B. The Intermediary's Problem

Proof of Proposition 1. The dynamic problem of the intermediary is

$$V(K_n, K_u) = \max_{K'_n, I} \left\{ R_n K_n + p_u R_u K_u - I + \frac{V(K'_n, K'_u)}{1+r} \right\}, \quad (\text{B1})$$

subject to

$$K'_n = (1 - \delta - \zeta)K_n + I, \quad K'_u = (1 - \delta - \kappa)K_u + \zeta K_n.$$

Take the first-order derivatives with respect to K_n and K_u :

$$\frac{\partial V(K_n, K_u)}{\partial K_n} = R_n + (1 - \delta - \zeta) + \frac{\partial V(K'_n, K'_u)}{\partial K'_u} \frac{\zeta}{1+r}, \quad (\text{B2})$$

$$\frac{\partial V(K_n, K_u)}{\partial K_u} = p_u R_u + \frac{\partial V(K'_n, K'_u)}{\partial K'_u} \frac{1 - \delta - \kappa}{1+r}. \quad (\text{B3})$$

The first-order condition with respect to K'_n is

$$\frac{\partial V(K_n, K_u)}{\partial K'_n} = -1 + \frac{\partial V(K'_n, K'_u)}{\partial K'_n} \frac{1}{1+r} = 0. \quad (\text{B4})$$

Equation (B3) implies

$$\frac{\partial V(K_n, K_u)}{\partial K_u} = p_u R_u + \frac{1 - \delta - \kappa}{1+r} p'_u R'_u + \left(\frac{1 - \delta - \kappa}{1+r} \right)^2 p''_u R''_u + \dots$$

At the steady state this is equivalent to

$$\frac{\partial V(K_n, K_u)}{\partial K_u} = p_u R_u \frac{1+r}{r + \delta + \kappa}. \quad (\text{B5})$$

Combine (B2) and (B4) to obtain

$$1+r = R_n + (1 - \delta - \zeta) + \frac{\zeta}{1+r} \frac{\partial V(K''_n, K''_u)}{\partial K''_u}. \quad (\text{B6})$$

Applying the steady-state condition (B5),

$$r = R_n - \delta - \zeta \left(1 - \frac{p_u R_u}{r + \delta + \kappa} \right), \quad (\text{B7})$$

which is (18). □

The split into a price and a user cost. Only the product $p_u R_u$ appears in (B7), so something more is needed to say what each factor is. Let p_u be the price a used unit actually trades at, the same price that is carried on the balance sheet (21) and in the financing constraint (9). Buy a unit today for p_u . Over the period it pays the rental $p_u R_u$, and at the end of the period what is left of it

is worth $(1 - \delta - \kappa)p_u$. The two together have to be worth $(1 + r)p_u$, or no one would hold it:

$$(1 + r)p_u = p_u R_u + (1 - \delta - \kappa)p_u.$$

This gives $R_u = r + \delta + \kappa$, and (B5) then reads $\partial V / \partial K_u = (1 + r)p_u$. So p_u is what a used unit is worth today with this period's rental still to come; once that rental has been paid, what is left is the smaller amount $(1 - \delta - \kappa)p_u$. Read the same line as $p_u R_u = r p_u + (\delta + \kappa)p_u$: the rental has to cover interest on what the unit is worth, its depreciation, and the chance that it breaks. That is why R_u is a user cost. Equation (B7) now reads $R_n = r + \delta + \zeta(p_n - p_u)$, and inverting it delivers (20). Only the product enters any firm's rental bill. The financing constraint (9), written on market value, is what makes the two factors matter separately.

Zero profit. The technology is constant returns and (B1) is linear, so V is homogeneous of degree one. By Euler's theorem, $V = V_n K_n^s + V_u K_u^s = (1 + r)(p_n K_n^s + p_u K_u^s)$. Deposits are the only liability and capital the only asset, which is (21). The intermediary therefore owes $(1 + r)D$ against a portfolio worth V , and profit is zero. This establishes the claim made in Section 5.4.

Appendix C. Derivations

C.1. The capital mix

Proof of Proposition 2(i). A physical unit of type j costs $p_j R_j$ to rent and absorbs p_j of financing capacity. At shadow cost $\mu \geq 0$ its effective price is therefore $p_j(R_j + \mu)$, and for a given level K of effective capital the entrepreneur solves

$$\min_{K_n, K_u} p_n(R_n + \mu)K_n + p_u(R_u + \mu)K_u \quad \text{s.t.} \quad \left[\phi^{1/\gamma} K_n^{\frac{\gamma-1}{\gamma}} + (1 - \phi)^{1/\gamma} K_u^{\frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}} = K. \quad (\text{C1})$$

The first-order conditions give the CES demand ratio (31), which factors as

$$\rho(\mu) = \bar{\rho} \left[\frac{R_n + \mu}{R_u + \mu} \right]^\gamma, \quad \bar{\rho} \equiv \frac{1 - \phi}{\phi} \left(\frac{p_n}{p_u} \right)^\gamma, \quad (\text{C2})$$

since the price ratio does not depend on μ . In particular $\rho^{uc} = \bar{\rho}(R_n/R_u)^\gamma$. Differentiating the bracket,

$$\frac{\partial}{\partial \mu} \frac{R_n + \mu}{R_u + \mu} = \frac{R_u - R_n}{(R_u + \mu)^2}, \quad (\text{C3})$$

so $\rho'(\mu) > 0$ if and only if $R_u > R_n$, which is (33). As $\mu \rightarrow \infty$ the bracket tends to one. Hence $\rho(\mu) \rightarrow \bar{\rho}$. $\square$

C.2. Collateral productivity

Proof of Proposition 2(ii). Normalize $K_n = 1$, so that $\rho = K_u$ and the bundle is $\Psi(\rho) = [\phi^{1/\gamma} + (1 - \phi)^{1/\gamma} \rho^{\frac{\gamma-1}{\gamma}}]^{\frac{\gamma}{\gamma-1}}$. The capacity absorbed is $p_n + p_u \rho$, so collateral productivity is $\Lambda(\rho) = \Psi(\rho)/(p_n + p_u \rho)$, as in (32). Maximizing Λ is equivalent to minimizing the collateral cost of one

unit of the bundle, $(p_n + p_u \rho)/\Psi(\rho)$. That function is strictly quasi-convex in ρ : for every $t > 0$ the map $\rho \mapsto p_n + p_u \rho - t\Psi(\rho)$ is strictly convex, since Ψ is strictly concave, so each sublevel set is an interval. Its minimizer is therefore unique, and the first-order condition returns the $\bar{\rho}$ of (C2): the mix that minimizes collateral at prices (p_n, p_u) is the same mix that (C2) approaches as $\mu \rightarrow \infty$, because the user costs drop out of the demand ratio in that limit. Hence

$$\bar{\Lambda} \equiv \Lambda(\bar{\rho}) = \left[\phi p_n^{1-\gamma} + (1-\phi) p_u^{1-\gamma} \right]^{1/(\gamma-1)}, \quad (\text{C4})$$

which is (34), and Λ is strictly increasing on $(0, \bar{\rho})$.

By (C2), $\rho(\mu)$ is $\bar{\rho}$ times a bracket that under (33) rises to one from below. Therefore $\rho(\mu)$ increases from $\rho^{uc} < \bar{\rho}$ toward $\bar{\rho}$, and $\Lambda(\rho(\mu))$ is strictly increasing in μ with limit $\bar{\Lambda}$. Finally $\bar{\Lambda} > p_n^{-1}$ whenever $p_u < p_n$ and $\gamma > 1$, since then $p_u^{1-\gamma} > p_n^{1-\gamma}$ and the convex combination in (C4) exceeds $p_n^{1-\gamma}$. $\square$

Two benchmarks should be distinguished. The p_n^{-1} of Proposition 2(ii) is collateral productivity in the economy without used capital, where $K = K_n$ and a dollar of collateral supports $1/p_n$ units of the bundle; it is the counterpart of the recalibrated no-used economy of Section 6. It is not $\Lambda(0) = \phi^{1/(\gamma-1)}/p_n$, the productivity of a firm that retains the share parameter $\phi < 1$ but is denied used capital. Since $\phi < 1$ and $\gamma > 1$, $\Lambda(0) < p_n^{-1}$, so the ratio $\bar{\Lambda}/\Lambda(0)$ combines the financing gain with a mechanical penalty from a missing CES input; $\bar{\Lambda} > \Lambda(0)$ holds by optimality alone and requires no condition on prices.

C.3. Which margin the relief runs on

Three price comparisons should be kept apart. Used capital is cheaper per physical unit, $p_u < p_n$, which by (20) is equivalent to $R_n > r + \delta$; this is what delivers $\bar{\Lambda} > p_n^{-1}$, and prices are the whole of that result. Its user cost per dollar of market value is higher, $R_u > R_n$ under (33); this is what delivers the tilt. Its rental payment per physical unit, $p_u R_u$, may lie on either side of $p_n R_n$, and no result here depends on which. Both $p_u < p_n$ and $R_u > R_n$ are equilibrium properties rather than restrictions on primitives, and both hold at the calibrated values.

The effective price $p_j(R_j + \mu)$ in (C1) already encodes the trade-off, and reading it per rental dollar makes the mechanism plain. A dollar of rental spending on type j hires $1/(p_j R_j)$ units and ties up

$$\frac{p_j}{p_j R_j} = \frac{1}{R_j} \quad (\text{C5})$$

of financing capacity. The price cancels, so on this margin the user cost alone decides: renting used capital frees collateral relative to new exactly when $R_u > R_n$, the equivalent forms of which are given in Proposition 2. What the constraint counts is market value, and a rental dollar spent on used capital commands less of it.

Appendix D. The Comparison Design and the Productivity Tail

The comparison in the text recalibrates the economy without used capital in full. Its Pareto-tail parameter therefore falls from 5.949 to 5.397. A fatter tail places more mass on highly productive entrepreneurs, who have the largest desired scale, and so raises measured misallocation on its own. This appendix holds the tail fixed in two ways. The *tail-pinned* economy keeps $(\eta_p, \omega, \sigma_y)$ at their benchmark values and recalibrates only (θ, β) . The *tail diagnostic* returns η_p to 5.949 in the fully recalibrated economy and re-clears prices without refitting any moment. Its targeted moments therefore no longer hold exactly. Misallocation reduced, $1 - \mathcal{M}_{\text{bm}}/\mathcal{M}$, is the share of that design’s TFP loss $\mathcal{M}$ —defined in (37)—that the benchmark reduces. Entry repaired is $1 - \mathcal{S}_{\text{bm}}/\mathcal{S}$, defined the same way from the entry shortfall $\mathcal{S}$ in (38). Each economy is compared with its own frictionless twin. Table 17 reports the benchmark and the three designs. Changing η_p alone moves the TFP loss

Table 17. The result under three designs without used capital

Economy	η_p	TFP loss $\mathcal{M}$ (log pts)	Misallocation reduced	Entry repaired	K/K^*	sd(log MRPK)
Benchmark (used K)	5.949	5.72	—	—	0.343	0.4501
No used capital, tail pinned	5.949	7.11	19.5%	32.0%	0.268	0.5298
Tail diagnostic	5.949	7.42	22.9%	27.8%	0.295	0.5296
No used capital, recalibrated	5.397	9.39	39.1%	22.6%	0.279	0.5977

from 7.42 to 9.39 log points. The collateral multiplier, productivity persistence, worker dispersion, the discount factor and equilibrium prices together account for 0.31. Under all three designs the benchmark has the lower TFP loss, the higher mean scale relative to target, the lower dispersion of log MRPK, and the higher rate of entry by poor talented agents. The share of the frictional loss attributed to used capital varies with the design. Its sign does not. The tail-pinned and tail-diagnostic economies report nearly identical sd(log MRPK) (0.5298 vs. 0.5296) because both use the benchmark productivity tail. They differ in $(\theta, \omega, \sigma_y, \beta)$, which move mean scale (K/K^* of 0.268 vs. 0.295) but barely move dispersion. Restoring η_p itself, from the recalibrated economy to the diagnostic, lowers sd(log MRPK) from 0.598 to 0.530. The two margins move in opposite directions. Misallocation reduced rises from 19.5 to 39.1 percent as the tail is refit, while entry repaired falls from 32.0 to 22.6. A fatter tail places more mass on very productive entrepreneurs, whose desired scale is largest, so constraining them is expensive and measured misallocation rises. But those same agents are inframarginal in the occupational choice: they are productive enough to enter even at a constrained scale, so the constraint blocks fewer of them and the entry shortfall is smaller.

Appendix E. Robustness to the Assigned CES Parameters

The weight ϕ on new capital and the elasticity of substitution γ in the capital aggregator (4) are assigned rather than calibrated; the benchmark sets $\phi = 0.70$ and $\gamma = 2.0$. Table 18 re-solves the benchmark at $\phi = 0.75$, the value used by Lanteri (2018), and at $\gamma \in \{1.7, 2.0, 2.5\}$, which brackets the range Edgerton (2011) estimates, in each case holding the calibrated parameters $(\theta, \eta_p, \omega, \kappa, \zeta, \sigma_y, \beta)$ at their benchmark values and re-clearing only the prices (W, r, R_n) . These are robustness sweeps rather than recalibrations: each asks whether the results turn on the assigned value, not whether the model could be re-fitted around it. The $\phi = 0.70$ and $\gamma = 2.0$ columns are the benchmark. Neither parameter matters for the conclusions: the TFP loss $\mathcal{M}$ stays between 5.70 and 5.75 log points, and misallocation reduced stays between 38.7 and 39.3 percent.

Table 18. Sensitivity of the results to the assigned CES parameters

	ϕ sweep		γ sweep		
	$\phi = 0.70$ (benchmark)	$\phi = 0.75$	$\gamma = 1.7$	$\gamma = 2.0$ (benchmark)	$\gamma = 2.5$
<i>Panel A: Gain from removing the collateral constraint</i>					
TFP (%)	5.89	5.91	5.92	5.89	5.86
TFP loss $\mathcal{M}$ (log points)	5.72	5.74	5.75	5.72	5.70
Misallocation reduced (%)	39.1	38.9	38.7	39.1	39.3
<i>Panel B: Allocation in the constrained equilibrium</i>					
Mean scale relative to target, K/K^*	0.343	0.350	0.347	0.343	0.337
Share of firms constrained	0.908	0.906	0.907	0.908	0.908
Entrepreneur share	0.221	0.220	0.221	0.221	0.220
<i>Panel C: Cleared prices</i>					
Wage W	0.540	0.542	0.540	0.540	0.540
Interest rate r	0.031	0.031	0.031	0.031	0.031
New-capital rental R_n	0.119	0.124	0.119	0.119	0.120
Used-capital price p_u	0.645	0.582	0.648	0.645	0.640

Notes: Each column re-clears (W, r, R_n) at the stated parameter value with the calibrated parameters held at their benchmark levels; the benchmark is $\phi = 0.70$, $\gamma = 2.0$. The $\phi = 0.70$ and $\gamma = 2.0$ columns refer to the same economy. $\mathcal{M}$ is the TFP loss from the collateral constraint defined in (37), measured against each economy's own frictionless counterpart; the TFP row is the same object in percent rather than log points.